Exploratory Data Analysis on

Strategic Portfolio Selection: Consumption Thematic Funds

by

Group 23

Akshada Modak
ID: 202301485
Course: BTech - ICT-CS

Maanav Gurubaxani
ID: 202301438
Course: Btech - ICT-CS

Khanak Patel
ID: 202303020
Course: BTech - MNC

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Under the guidance of

Dr. Gopinath Panda

Dhirubhai Ambani University

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Acknowledgment

I am writing this letter to express my heartfelt gratitude for your guidance and support throughout the duration of my project titled “[Project Title].” Your invaluable assistance has played a pivotal role in shaping the successful completion of this endeavor.

I am extremely fortunate to have had the opportunity to work under your mentorship. Your expertise, encouragement, and willingness to share your knowledge have been instrumental in elevating the quality and scope of my project. Your constructive feedback and insightful suggestions have helped me overcome challenges and develop a deeper understanding of the subject matter.

Furthermore, I would like to extend my appreciation to the entire team at [Organization/Institution Name] for fostering an environment of collaboration and innovation. The resources and facilities provided have been crucial in conducting comprehensive research and analysis.

I would also like to express my gratitude to my peers and colleagues who have been supportive throughout this journey. Their valuable input and camaraderie have been a constant source of motivation.

Completing this project has been a tremendous learning experience, and I am confident that the knowledge and skills acquired during this endeavor will serve as a solid foundation for my future endeavors.

Once again, thank you for your unwavering guidance and belief in my abilities. Your mentorship has been invaluable, and I am truly grateful for the opportunity to work with you.

Sincerely,
[Akshada Modak, 202301485] [Maanav Gurubaxani, 202301438] [Khanak Patel, 202303020]

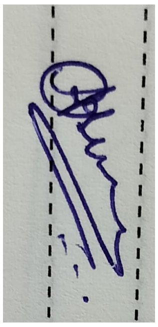
DECLARATION

We, Akshada Modak, Maanav Gurubaxani and Khanak Patel hereby declare that the EDA project work presented in this report is our original work and has not been submitted for any other academic degree. All the sources cited in this report have been appropriately referenced.

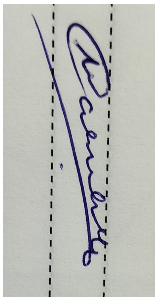
We acknowledge that the data used in this project is obtained from the AMFI (www.amfiindia.com) site. We also declare that we have adhered to the terms and conditions mentioned in the website for using the dataset. We confirm that the dataset used in this project is true and accurate to the best of our knowledge.

We acknowledge that we have received no external help or assistance in conducting this project, except for the guidance provided by our mentor Prof. Gopinath Panda. We declare that there is no conflict of interest in conducting this EDA project.

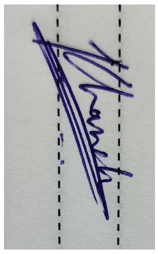
We hereby sign the declaration statement and confirm the submission of this report on 2nd July, 2023.



Akshada Modak
ID: 202301485
Course: BTech - ICT-CS



Maanav Gurubaxani
ID: 202301438
Course: Btech - ICT-CS



Khanak Patel
ID: 202303020
Course: BTech - MNC

CERTIFICATE

This is to certify that Group 23 comprising Akshada Modak, Maanav Gurubaxani and Khanak Patel has successfully completed an exploratory data analysis (EDA) project on the Strategic Portfolio Selection: Consumption Thematic Funds which was obtained from www.amfiindia.com

The EDA project presented by Group 23 is their original work and has been completed under the guidance of the course instructor, Prof. Gopinath Panda, who has provided support and guidance throughout the project. The project is based on a thorough analysis of the AMFI dataset, and the results presented in the report are based on the data obtained from the dataset.

This certificate is issued to recognize the successful completion of the EDA project on the Strategic Portfolio Selection: Consumption Thematic Funds which demonstrates the analytical skills and knowledge of the students of Group 23 in the field of data analysis.

Signed,
Dr. Gopinath Panda,
IT 462 Course Instructor
Dhirubhai Ambani Institute of Information and Communication Technology
Gandhinagar, Gujarat, INDIA.

December 1, 2025

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Abstract

This project conducts a comprehensive Exploratory Data Analysis (EDA) of India's Consumption Thematic Mutual Funds using seven years of daily NAV, AUM, and benchmark data sourced from AMFI. The analysis examines long-term performance, risk behavior, and manager skill through metrics such as CAGR, volatility, Sharpe and Sortino ratios, maximum drawdown, and CAPM-based alpha and beta. Extensive visualizations and statistical tests reveal that while certain schemes deliver strong risk-adjusted performance, most funds exhibit high correlation, limited diversification benefits, and no statistically significant alpha generation. Additional multivariate analyses—including market regime classification, AUM-performance relationships, ranking persistence, and predictive modeling for volatility and drawdowns—provide deeper insight into fund behavior across market conditions. Overall, the study offers a data-driven evaluation of consumption-focused portfolios, highlighting the drivers of performance and the constraints of active management within this thematic category.

Chapter 1. Introduction

1.1 Your Project Idea

The Indian mutual fund industry has witnessed rapid growth in thematic investment categories, with consumption-focused funds emerging as a major area of interest. These funds invest in companies that benefit from rising consumer demand, lifestyle changes, and structural economic growth.

The goal of this project is to perform an in-depth Exploratory Data Analysis (EDA) on **seven years of historical data (2019–2025)** for major Consumption Mutual Fund schemes in India. The analysis aims to uncover meaningful insights related to risk, performance, alpha generation, and asset growth. Specifically, the objectives of this project are:

1. **Identify top-performing schemes** based on long-term risk-adjusted returns.
2. **Analyze risk drivers** such as volatility, maximum drawdown, and downside deviation.
3. **Evaluate performance metrics** including CAGR, Sharpe Ratio, and Sortino Ratio across schemes.
4. **Study alpha generation** using CAPM-based alpha, beta, and R-squared values.
5. **Assess AUM dynamics** including AUM growth, fund flows, and market share changes.
6. **Predict key outcomes** such as volatility and maximum drawdown using machine learning models.
7. **Conduct regime analysis** to evaluate how funds perform during different market phases.

This project provides a comprehensive understanding of Consumption Mutual Funds, helping investors, analysts, and policymakers make informed decisions based on quantitative insights.

1.2 Data Collection

Our analysis is based on daily scheme–level data obtained from the official website of the Association of Mutual Funds in India (AMFI). The AMFI "Fund Performance" section allows users to filter mutual funds by type, category, and date, and then view or download the corresponding scheme–wise performance table.

Figure 1.1 shows the main fund performance page with filters for scheme type (open–ended), asset class (equity), and category (sectoral / thematic). Once the desired filters and report date are selected, the website returns a tabular view of all matching schemes, including their riskometer labels, latest NAV, multi–year return metrics, and AUM.

After selecting the appropriate filters, the site displays a detailed table of consumption–oriented schemes, as illustrated in Figure 1.2. This table contains, for each scheme, its benchmark index, riskometer classification, latest NAV, multi–horizon return statistics, information ratios, and daily AUM. While the user interface allows manual download for a single date, our analysis requires a continuous daily time series across several years.

To obtain historical data efficiently, we inspected the browser's developer tools while interacting with the AMFI website. As shown in Figure 1.3, each time the user requests fund performance for a specific date, the front–end sends a POST request to an internal API endpoint:

`https://www.amfiindia.com/gateway/pollingsebi/api/amfi/fundperformance`

with a JSON payload that encodes the selected filters, including the report date. This observation allowed us to bypass manual downloads and instead construct a reliable web–scraping pipeline that queries the same endpoint programmatically for each day in our analysis period.

The final scraping script iterates from the start of 2019 up to the most recent available date. For each day, it sends a POST request to the AMFI endpoint with a payload that fixes the fund category to sectoral/thematic equity (consumption funds) and varies only the `reportDate` field. The script parses the JSON response into a pandas DataFrame, appends the resulting records to an in–memory list, and tags every row with the corresponding report date.

To make the scraping robust, we incorporate:

- **Incremental saving** every month,
- **Automatic resume** from last saved date,
- **Error handling** for API timeouts.

Listing 1.1: AMFI Daily Data Scraper for 7 Years

```python
import requests
import pandas as pd
from datetime import date, timedelta
import time
import os

# --- Configuration ---
API_URL = "https://www.amfiindia.com/gateway/pollingsebi/api/amfi/
    fundperformance"
START_DATE = date(2019, 1, 1)
END_DATE = date.today()
OUTPUT_FILE = "amfi_thematic_performance_master.csv"
TEMP_FILE = "temp_master_data.csv" # Temporary file for saving
    progress
MONTHLY_SAVE_POINT = START_DATE.month

# Fixed payload parameters for Consumption Funds
BASE_PAYLOAD = {
    "maturityType": 1,
    "category": 1,
    "subCategory": 12,
```

```python
    "mfid": 0,
    "reportDate": ""
}

HEADERS = {
    'Content-Type': 'application/json',
    'User-Agent': 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36'
}

all_data = []
current_date = START_DATE

print(f"Starting data extraction from {START_DATE.strftime('%Y-%m-%d')}...")

# --- Resume if a temporary file exists ---
if os.path.exists(TEMP_FILE):
    print(f"Resuming from existing temporary file: {TEMP_FILE}")
    try:
        master_df = pd.read_csv(TEMP_FILE)
        last_date_str = master_df['Report_Date'].max()
        if last_date_str:
            last_date = pd.to_datetime(last_date_str, format='%d-%b-%Y').date()
            current_date = last_date + timedelta(days=1)
            print(f"Resuming from date: {current_date.strftime('%Y-%m-%d')}")
        all_data = [master_df]
    except Exception as e:
        print(f"Could not load temporary file: {e}. Starting fresh.")

# --- Daily scraping loop ---
while current_date <= END_DATE:
    date_str = current_date.strftime("%d-%b-%Y")
    BASE_PAYLOAD['reportDate'] = date_str

    print(f"Processing: {date_str}...", end='\r')

    try:
        response = requests.post(API_URL, json=BASE_PAYLOAD, headers=HEADERS, timeout=15)

        if response.status_code == 200:
            json_data = response.json()

            if json_data.get('validationStatus') == 'SUCCESS' and json_data.get('data'):
                df = pd.DataFrame(json_data['data'])
```

```
                df['Report_Date'] = date_str
                all_data.append(df)

                # Monthly checkpoint
                if current_date.month != MONTHLY_SAVE_POINT:
                    master_df = pd.concat(all_data, ignore_index=True)
                    master_df = master_df.drop_duplicates(
                        subset=['schemeName', 'navDate', 'Report_Date'
                            ]
                    )
                    master_df.to_csv(TEMP_FILE, index=False)
                    all_data = [master_df]
                    MONTHLY_SAVE_POINT = current_date.month
                    print(f"\nProgress saved up to {date_str}.")

    except requests.exceptions.RequestException as e:
        print(f"\nConnection error on {date_str}: {e}")

    current_date += timedelta(days=1)
    time.sleep(1)

# --- Final save ---
if all_data:
    master_df = pd.concat(all_data, ignore_index=True)
    master_df = master_df.drop_duplicates(
        subset=['schemeName', 'navDate', 'Report_Date']
    )
    master_df.to_csv(OUTPUT_FILE, index=False)
    if os.path.exists(TEMP_FILE):
        os.remove(TEMP_FILE)

    print("\n========================================")
    print("Success! MASTER DATA FILE CREATED.")
    print(f"Total Unique Records: {len(master_df)}")
    print(f"Saved to: {OUTPUT_FILE}")
    print("========================================")
else:
    print("\nNo data was collected.")
```

		Riskometer		Latest NAV (₹)		5-Year Return (%)			5-Year Information Ratio		
Scheme Name	Benchmark	Scheme	Benchmark	Regular	Direct	Regular	Direct	Benchmark	Regular	Direct	Daily AUM (Cr.)
360 ONE Quant Fund	BSE 200 TRI	Very High	Very High	19.7610	20.7765	--	--	--	--	--	904.04
Aditya Birla Sun Life Banking & Financial Services Fund	Nifty Financial Services TRI	Very High	Very High	64.6500	72.7600	16.74	17.87	15.34	0.21	0.38	3714.35
Aditya Birla Sun Life Business Cycle Fund	BSE 500 TRI	Very High	Very High	15.6000	16.4400	--	--	--	--	--	1802.63
Aditya Birla Sun Life Consumption Fund	Nifty India Consumption TRI	Very High	Very High	220.5700	253.1900	17.60	18.92	19.51	-0.33	-0.11	6534.64
Aditya Birla Sun Life Digital India	BSE Teck TRI	Very High	Very High	177.1600	200.0900	17.86	19.24	14.97	0.47	0.69	4858.83

Figure 1.1: Fund performance page on the AMFI website with filters for mutual fund schemes.

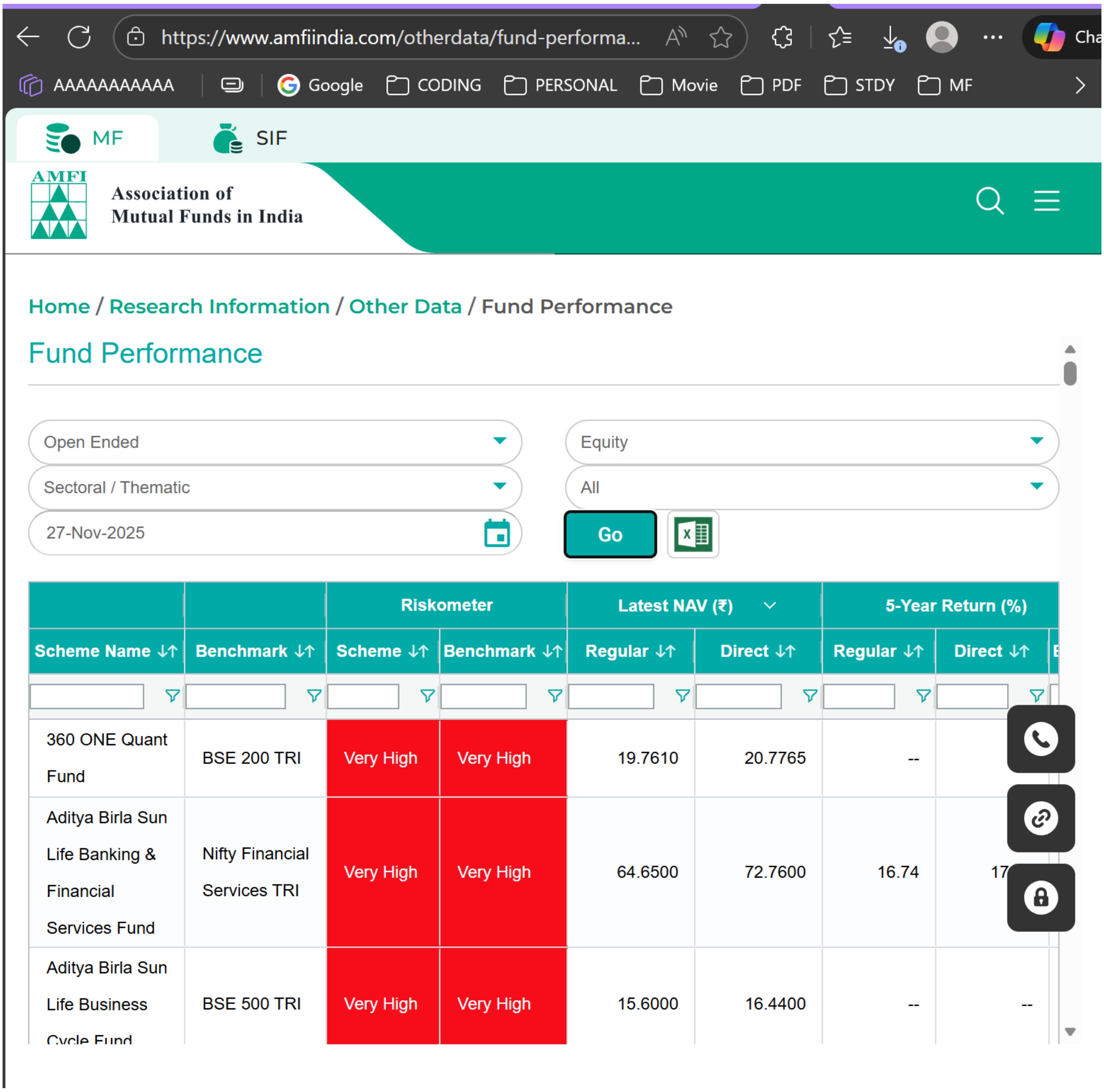

Figure 1.2: Tabular view of scheme performance, including riskometer, NAV, returns, and AUM.

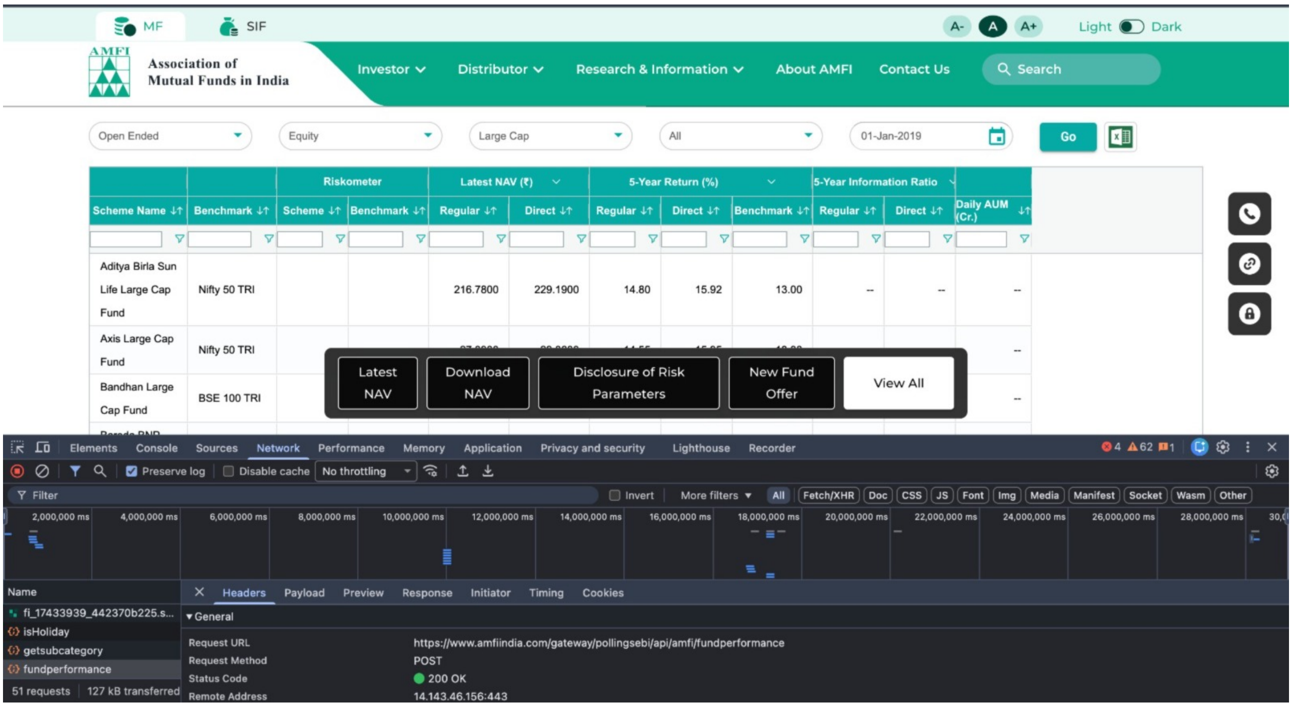

Figure 1.3: Browser Network panel capturing the `fundperformance` API request used for scraping.

1.3 Dataset Description

The scraping pipeline described above produces a consolidated dataset in *long* format, where each row corresponds to a unique combination of trading date and mutual fund scheme. Every observation therefore represents the state of a given consumption fund on a particular day, together with its associated performance metrics and attributes.

The key characteristics of the dataset are summarised below:

- **Time period:** The dataset spans from 1 January 2019 to the most recent available date in 2025, providing roughly seven years of daily history for thematic consumption funds.
- **Schemes:** It covers multiple open–ended equity schemes that fall under the consumption or similar sectoral/thematic categories as defined on the AMFI platform. Each scheme appears repeatedly across days, forming a time series of observations.
- **NAV information:** For each scheme and date, we collect the latest Net Asset Value (NAV) for both regular and direct plans. This forms the primary price series used to compute derived performance and risk metrics.
- **Return metrics:** The dataset includes returns over multiple horizons such as 7-day, 15-day, 1-month, 3-month, 6-month, 1-year, 3-year, 5-year, 10-year, and since–launch returns (where available). Having this panel of horizons allows us to study both short–term and long–term behaviour of each fund.
- **Benchmark information:** For every scheme, the AMFI table reports the corresponding benchmark index and its return metrics over the same horizons. These benchmark returns are essential for computing relative performance measures such as excess returns, beta, and CAPM alpha.

- **Risk and classification attributes:** The dataset records riskometer labels for the scheme and its benchmark (for example, "Very High"), along with other scheme–level descriptors that help us categorise funds consistently across time.
- **Assets under management (AUM):** Each row also contains the scheme's daily AUM in crore rupees. This allows us to analyse how asset size and investor flows evolve jointly with performance and risk.

Because the data is organised in long format, it is naturally suited for time-series analysis and panel-data style comparisons across schemes. In later chapters, we will transform this raw dataset into a set of cleaned and derived features, compute risk and performance statistics at different horizons, and construct target variables for our predictive models.

1.4 Packages Required

The project uses a comprehensive set of Python libraries for data ingestion, cleaning, statistical analysis, visualization, machine learning, and model evaluation. Table 1.1 lists all required packages used throughout the analysis.

Package	Purpose
`warnings`	Suppress non-critical warnings
`pandas`	Data manipulation and analysis
`numpy`	Numerical computations
`matplotlib`	Plotting and visualizations
`seaborn`	Statistical visualizations
`pathlib`	File system path handling
`json`	JSON data processing
`datetime`	Date/time manipulation
`statsmodels`	Statistical tests and regression models
`scipy`	Statistical functions and probability tests
`sklearn.model_selection`	Train-test split, cross-validation
`sklearn.preprocessing`	Scaling, label encoding
`sklearn.ensemble`	Random Forest models
`sklearn.linear_model`	Linear and logistic regression
`sklearn.metrics`	Regression and classification metrics
`xgboost`	Gradient boosting models (XGBRegressor, XGBClassifier)
`joblib`	Saving/loading trained models
`numpy.random`	Reproducibility with fixed seed

Table 1.1: List of Python packages required for the project.

All plots use a consistent visual theme by configuring Matplotlib and Seaborn styling parameters for clarity and publication-ready quality.

Chapter 2. Data Cleaning

2.1 Missing Data Analysis

To ensure high-quality inputs for statistical modeling and machine learning, a thorough missing data analysis was performed on the raw dataset loaded from `ULTIMATE.csv`. The dataset contains daily NAV, AUM, scheme information, and derived fields for all thematic consumption funds from 2019–2025.

2.1.1 Dataset Overview

The dataset was first inspected to understand its structure:

- **Total rows:** The dataset contains a large number of records representing daily observations.
- **Unique Schemes:** Each mutual fund scheme was counted, and the number of available records per scheme was displayed.
- **Date Validation:** The variable `navDate` was converted to a standard date format using `pd.to_datetime`, allowing for verification of the full date range.

A small excerpt of the raw dataset (first 5 rows) was printed to inspect key fields such as `navDate`, `schemeName`, `navDirect`, `navRegular`, and `dailyAUM`.

2.1.2 Computation of Missing Data

For each column, the following were computed:

- Total number of missing values.
- Percentage of missing values relative to total rows.

The results were compiled into a sorted table, highlighting columns with non-zero missing values. Columns with the highest missing counts (up to top 20) were further inspected.

If missing values existed, the analysis also quantified:

- Number of columns with $> 50\%$ missing data.
- Number of columns with $> 90\%$ missing data.

2.1.3 Visualization

To better understand the distribution of missing data across columns, a bar chart was generated showing the top 20 columns by missing percentage. The plot was exported as:

`missing_data_analysis.png`

The figure can be included in the report as shown below:

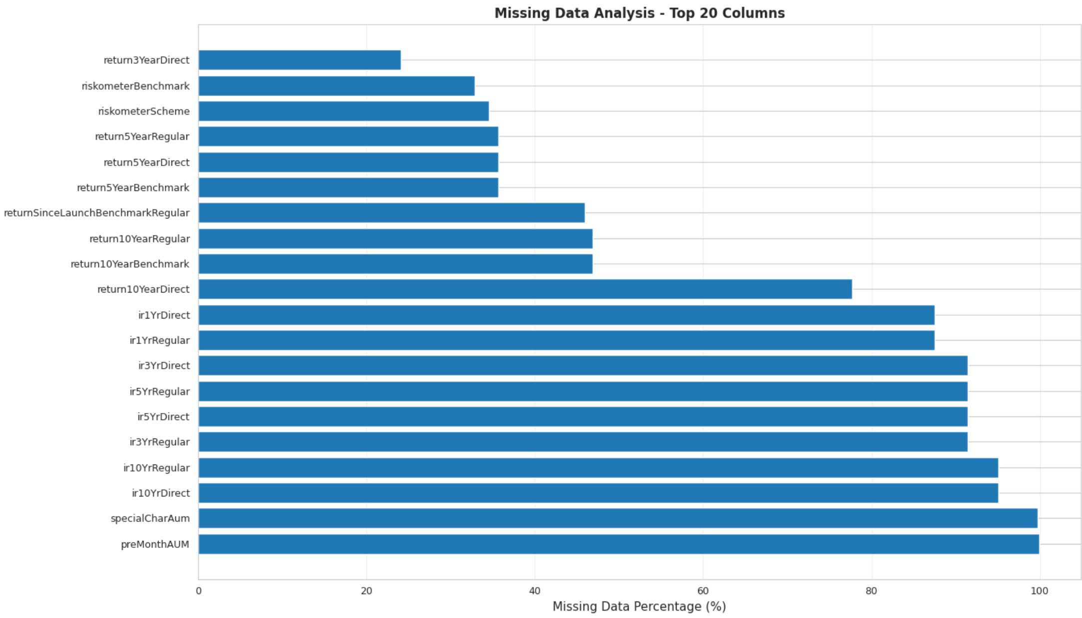

Figure 2.1: Missing Data Analysis (Top 20 Columns)

This analysis helps identify variables requiring imputation, removal, or further preprocessing before model development.

2.2 Imputation

After identifying missing values in NAV and AUM fields, an imputation strategy was implemented to ensure continuity in the time series and preserve the integrity of scheme-wise analysis.

2.2.1 Forward Filling Within Each Scheme

For each mutual fund scheme, the following columns were imputed using **forward fill (ffill)**:

- `NAV Direct`
- `NAV Regular`
- `Daily AUM (Cr.)`

Forward fill ensures that the most recent available value is carried forward, which is appropriate for financial time-series where NAV and AUM typically change gradually. The imputation was performed *scheme-wise* to prevent leakage across different funds.

2.2.2 Handling Remaining Missing Values

After forward filling, rows that still contained missing values in essential fields were removed. Specifically, any record missing:

- `NAV Direct`, or
- `NAV Date`

was dropped from the dataset.

This step ensures that all remaining observations are complete and usable for downstream analysis, including return computation, rolling metrics, and modeling.

2.2.3 Post-Imputation Summary

The following information was tracked during the process:

- Number of values forward-filled for each NAV and AUM column.
- Final dataset shape after removing irreparable missing values.
- Percentage of rows dropped due to missing essential data.
- Remaining missing values, if any, in NAV or AUM fields.

If any key columns still contained missing data, their counts and percentages were reported. In most cases, the imputation process successfully resolved all missing values for essential fields.

2.2.4 Result

The final cleaned dataset contained no missing values in:

- `NAV Direct`
- `NAV Regular`
- `Daily AUM (Cr.)`

This ensures a fully consistent dataset for visualization, feature engineering, statistical testing, and machine learning model development.

Chapter 3. Visualization

3.1 Univariate Analysis

This section presents the univariate analysis of all performance and risk metrics computed from the NAV (Net Asset Value) time-series of mutual funds. Each metric is analysed individually to understand its distribution, central tendency, and variability.

3.1.1 Compound Annual Growth Rate (CAGR)

Formula:

$$CAGR = \left(\frac{Ending\,Value}{Beginning\,Value}\right)^{1/Years} - 1$$

Interpretation: CAGR measures the annualized growth rate of a mutual fund assuming smooth, consistent growth.

Importance:

- Represents long-term performance.
- Smooths out volatility.
- Higher CAGR indicates superior growth.

3.1.2 Annualized Volatility

Formula:

$$Volatility = StdDev(daily\,returns) \times \sqrt{252}$$

Interpretation: Measures the overall risk of the fund by evaluating how widely daily returns fluctuate.

Importance:

- High volatility = unstable/ risky.
- Low volatility = stable returns.
- Used in Sharpe and Sortino ratios.

3.1.3 Sharpe Ratio

Formula:

$$Sharpe = \frac{CAGR - Risk\ Free\ Rate}{Volatility}$$

Interpretation: Indicates excess return generated per unit of total risk.
Scale:

- > 1.0: Good
- 0.5 to 1.0: Moderate
- < 0.5: Poor

3.1.4 Sortino Ratio

Formula:

$$Sortino = \frac{CAGR - Risk\ Free\ Rate}{Downside\ Volatility}$$

Interpretation: Measures risk-adjusted return but only penalizes downside volatility.

3.1.5 Maximum Drawdown (MDD)

Formula:

$$MDD = \min\left(\frac{NAV}{Rolling\ Peak} - 1\right)$$

Interpretation: Represents the largest fall from a peak, indicating downside risk and worst-case scenario.

3.1.6 Average AUM (Assets Under Management)

Meaning: The average total value of assets managed by the fund.
Importance:

- Indicates trust and liquidity.
- Large AUM = lower liquidity risk.

3.1.7 Data Points and Years Covered

Meaning:

- Number of NAV entries available.
- Number of years between first and last NAV value.

Importance:

- More data points = more reliable metrics.
- Longer time horizon improves trend detection.

3.1.8 NAV Evolution Over Time

To understand the long-term behavior of different mutual fund schemes, the daily **NAV (Net Asset Value)** values were analyzed as a univariate time series. Since different funds started at different NAV levels, a normalized NAV was computed by scaling all NAV series to a common base value of 100. This ensures a fair comparison of growth trends.

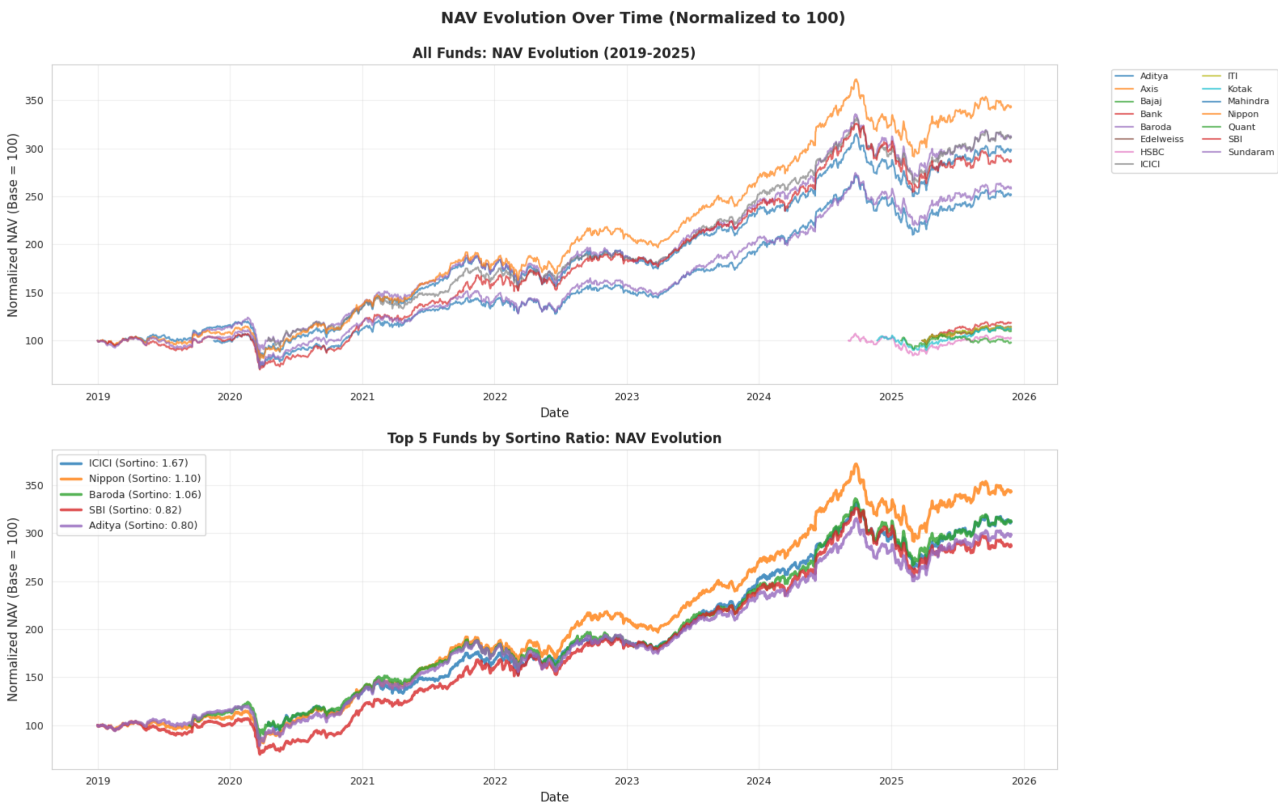

Figure 3.1: NAV Evolution of Mutual Funds (Normalized to 100)

The first plot shows the NAV trajectories of all funds between 2019–2025, reflecting their overall growth patterns, volatility, and long-term trends. The second plot highlights the **Top 5 funds ranked by the Sortino Ratio**, revealing much smoother and steeper upward trajectories compared to the average fund.

3.1.9 Cumulative Returns Analysis

Apart from visual NAV progression, the following univariate metrics were computed for each fund:

- **Total Return (%)** — Percentage growth from the first to last NAV.
- **Annualized Return (%)** — CAGR based on the full time duration.
- **Years Covered** — Total NAV history available for that scheme.

These metrics help quantify absolute and annualized performance for each scheme.

Table 3.1: Cumulative Returns Summary of Mutual Funds

Scheme Name	Total Return (%)	Annualized Return (%)	Years
Bank of India Consumption Fund	18.42	28.66	0.67
ICICI Prudential Bharat Consumption Fund	211.50	22.41	5.62
Edelweiss Consumption Fund	14.41	22.22	0.67
ITI Bharat Consumption Fund	13.87	21.36	0.67
Baroda BNP Paribas India Consumption Fund	211.62	20.08	6.21
Nippon India Consumption Fund	242.94	19.54	6.90
Axis Consumption Fund	12.22	18.75	0.67
Aditya Birla Sun Life Consumption Fund	198.05	17.14	6.90
Mahindra Manulife Consumption Fund	151.51	16.54	6.03
SBI Consumption Opportunities Fund	186.19	16.45	6.90
Bajaj Finserv Consumption Fund	9.71	14.81	0.67
Sundaram Consumption Fund	158.02	14.71	6.90
Kotak Consumption Fund	12.09	11.93	1.01
HSBC Consumption Fund	2.58	2.09	1.23
Quant Consumption Fund	-1.95	-2.34	0.83

Funds such as **ICICI Prudential**, **Nippon India**, and **Baroda BNP Paribas** demonstrate strong long-term annualized returns over multi-year periods. On the other hand, funds like **HSBC** and **Quant** exhibit weaker or negative annualized performance, as seen in their relatively flat or declining NAV patterns.

3.1.10 Risk Metrics Distribution Analysis

To further evaluate the risk characteristics of the consumption-based mutual funds, four key risk metrics were analyzed using univariate distribution plots: **Volatility**, **Maximum Drawdown**, **Sortino Ratio**, and **Sharpe Ratio**. These metrics help quantify downside behavior, consistency, and risk-adjusted performance.

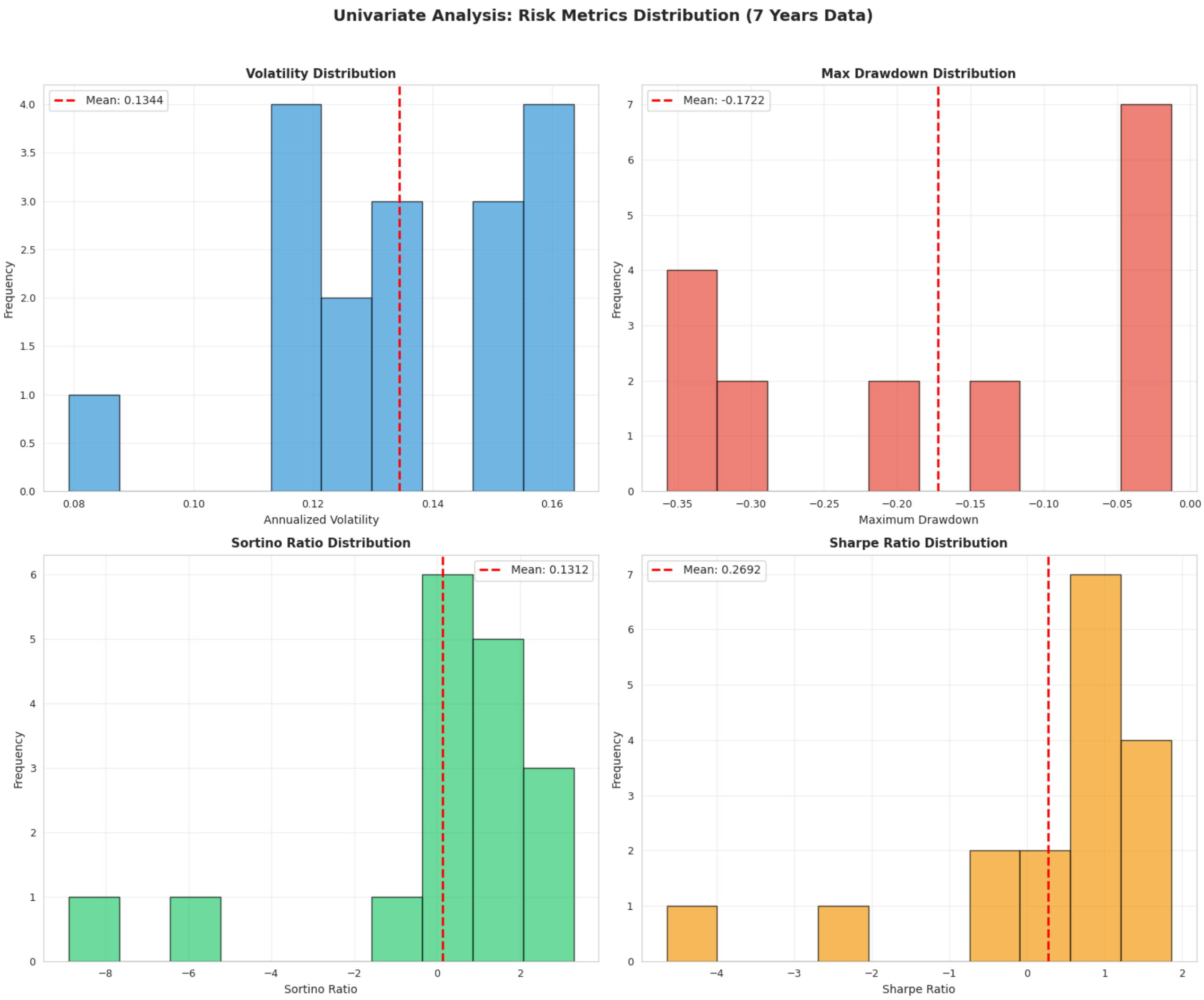

Figure 3.2: Distribution of Risk Metrics: Volatility, Max Drawdown, Sortino Ratio, Sharpe Ratio

The distributions and summary statistics reveal important insights into the risk behavior of consumption-oriented mutual funds.

1. Volatility Distribution

- Volatility ranges between **0.079 and 0.163**, with an average of **0.1344**.
- Most funds lie between **0.11 and 0.16**, indicating moderate and consistent risk levels.
- The distribution appears tight, suggesting similar volatility characteristics across funds.

Conclusion: Consumption funds generally exhibit **moderate and stable volatility** with no extreme outliers.

2. Maximum Drawdown Distribution

- Maximum drawdown ranges from **-0.357 (worst)** to **-0.013 (best)**.

- The average drawdown is **-0.172**, indicating that funds typically saw a 17% peak-to-trough drop over seven years.
- The distribution is wide and skewed, showing significant variation in downside protection.

Conclusion: There is substantial variability in downside risk—some funds offer strong drawdown protection while others faced **deep declines during market corrections**.

3. Sortino Ratio Distribution

- Sortino Ratio values vary widely between **-8.88 and 3.28**.
- The mean is **0.13**, but a high standard deviation indicates extreme variation.
- Several funds show **negative Sortino Ratios**, meaning downside volatility exceeds their excess returns.
- A few top performers have **Sortino** $>$ **2**, indicating exceptional downside protection.

Conclusion: Downside performance is inconsistent. While some funds offer strong risk-adjusted returns, many struggle with downside volatility.

4. Sharpe Ratio Distribution

- Sharpe Ratio ranges from **-4.64 to 1.85**.
- The average value is **0.269**, indicating modest category-wide efficiency.
- Several funds show negative Sharpe values, meaning poor returns relative to risk.
- A few achieve **Sharpe** $>$ **1**, signaling strong risk-adjusted performance.

Conclusion: Risk-adjusted performance varies significantly across the category, with only a small number of funds demonstrating consistently strong efficiency.

Overall Insights

- Volatility remains **consistent**, but downside risks (drawdowns) vary widely.
- Sortino and Sharpe ratios are highly dispersed, indicating large differences in risk-adjusted performance.
- Some funds show strong, stable, downside-protected returns, while others demonstrate weak risk control.

Final Summary:
The univariate analysis of risk metrics indicates that while consumption funds share similar volatility profiles, they differ markedly in **downside behavior** and **risk-adjusted performance**. Fund selection becomes crucial due to these wide variations.

3.1.11 Market Regime Analysis (Bull vs Bear Markets)

With 7 years of NAV data (2019–2025), it is possible to identify different market regimes and analyze fund performance during **Bull**, **Bear**, and **Neutral** periods.

Market Regime Detection

Market regime detection is a technique used to classify the market into different phases using historical return patterns. This helps understand how the overall market environment influenced mutual fund performance.

Methodology:

1. Compute the **average NAV** across all funds for each day.
2. Calculate **daily market returns** from this average NAV.
3. Compute **rolling 1-year returns (252 trading days)** to smooth short-term fluctuations.
4. Classify periods as:
    - **Bull Market:** Rolling return $> +10\%$
    - **Bear Market:** Rolling return $< -10\%$
    - **Neutral Market:** Rolling return between -10% and $+10\%$

This produces a **time series of market regimes** showing periods of market growth, decline, or stability.

Thresholds and Rolling Returns

- **Bull Threshold:** +10%
- **Bear Threshold:** -10%

Rolling Returns (First 20 Observations):

```
NAV Date
2020-01-10   -0.209931
2020-01-13   -0.195916
2020-01-14   -0.186625
2020-01-15   -0.180854
2020-01-16   -0.176876
2020-01-17   -0.172236
2020-01-20   -0.178180
2020-01-21   -0.184280
2020-01-22   -0.183637
2020-01-23   -0.174899
2020-01-24   -0.173655
2020-01-27   -0.174162
```

```
2020-01-28    -0.178746
2020-01-29    -0.171045
2020-01-30    -0.177048
2020-01-31    -0.175802
2020-02-03    -0.178163
2020-02-04    -0.162895
2020-02-05    -0.145843
2020-02-06    -0.136593
```

Table 3.2: Rolling 1-Year Market Returns Summary (2019–2025)

	count	mean	std	min	50%	max
NAV Direct	1460	1.565	1.277	-0.723	1.953	4.903

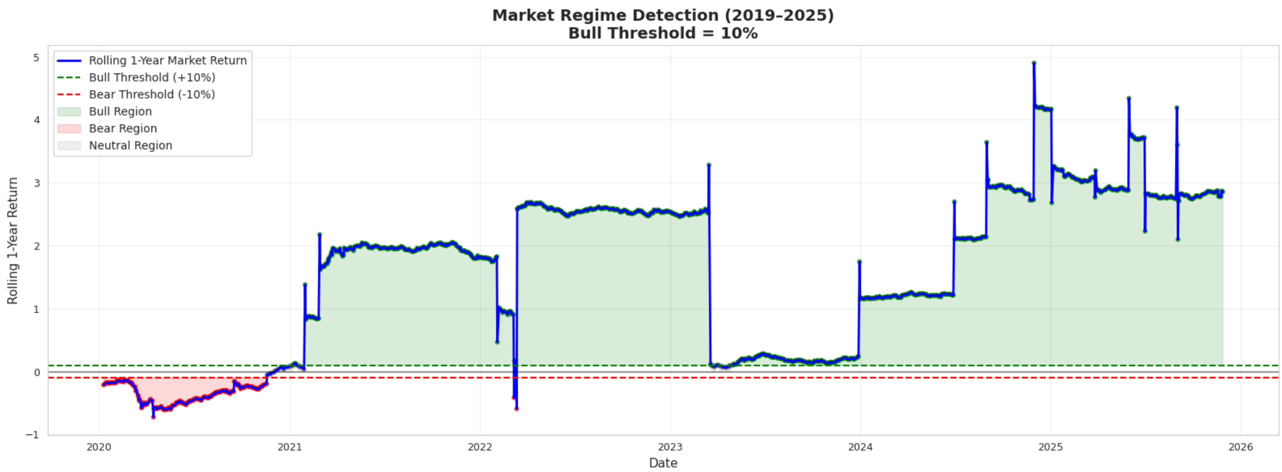

Figure 3.3: Distribution of Risk Metrics: Volatility, Max Drawdown, Sortino Ratio, Sharpe Ratio

Rolling Returns Summary Statistics:

Market Regime Statistics

Table 3.3: Market Regime Classification (2019–2025)

Regime	Days Count	Percentage (%)
Bull	1183	81.0
Bear	216	14.8
Neutral	61	4.2

Interpretation:

- The market was in a **Bull phase** for most of the period (81%), reflecting strong growth in the consumption fund sector.

- **Bear periods** were less frequent (14.8%), indicating short-term corrections.
- **Neutral periods** were rare (4.2%), showing that markets mostly exhibited a clear trend rather than stability.

This regime analysis provides context for evaluating fund performance under different market conditions.

3.1.12 Yearly Market vs Individual Fund Performance

To complement the market regime analysis, we visualize the **annual returns of individual consumption funds relative to the overall market** for each year from 2019 to 2024. The market return is computed as the average NAV across all funds.

Figure 3.4: Annual Returns of Individual Funds Compared to Market Average (2019–2024). Hatched bars indicate the market return; green bars represent positive returns, red bars represent negative returns.

Interpretation:

- **Market Bar (hatched):** Shows the overall market return for the year. Colors indicate regime:
  - Green: Bull Market (return $> 10\%$)
  - Red: Bear Market (return $< -10\%$)
  - Gray: Neutral Market (-10% $\leq$ return $\leq$ 10%)
- **Individual Fund Bars:** Show each fund's annual return. Green = positive, Red = negative.
- **Zero Line:** Indicates 0% return for reference.
- **Market Return Line (dashed blue):** Highlights the annual market return to compare fund performance.

Observations:

1. **Market Performance:** Bull years exhibit market bars above 0%, Bear years below 0%, and Neutral years near 0%.
2. **Fund vs Market Comparison:** Some funds consistently outperform the market (bars above the market line), while others underperform.
3. **Year-to-Year Volatility:** Certain funds show more volatile returns than the market, reflecting fund-specific risk.
4. **Investor Insights:** Consistent outperformers across multiple years are strong candidates for long-term investment.

Conclusion: The year-by-year visualization provides a comprehensive view of fund performance under different market regimes, allowing investors to identify **winners, laggards, and market trends**. It underscores the importance of **regime awareness**, as fund behavior varies significantly in Bull, Bear, and Neutral years.

3.2 Multivariate Analysis

This section examines the relationships between different performance and risk metrics of consumption funds, focusing on **Risk vs Performance**, **Alpha**, and **AUM**. The primary priority is understanding the risk-return landscape over the last 7 years.

3.2.1 Risk vs Performance

Two scatter plots are used to visualize the **risk-return profile** of all funds:

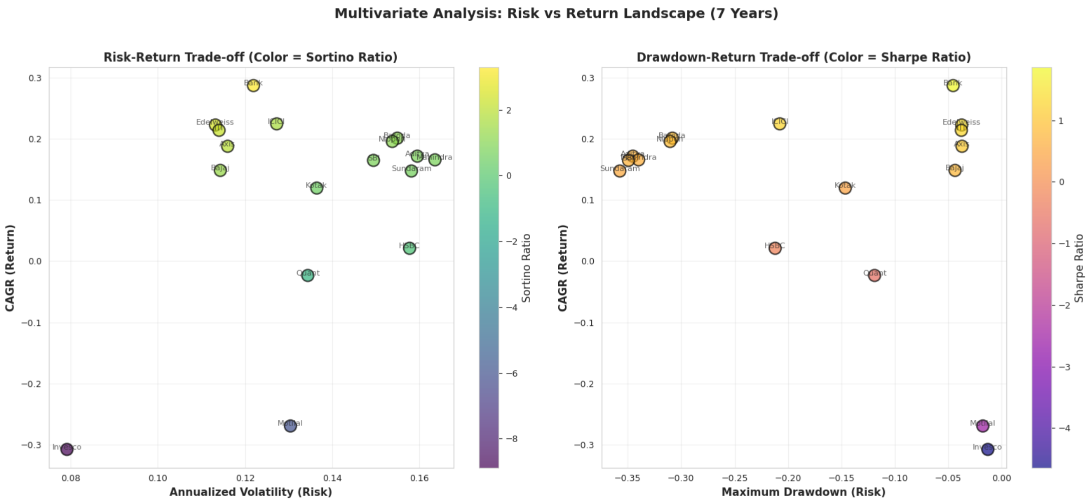

Figure 3.5: Risk-Return Analysis of Mutual Funds (2019–2025). Left: Volatility vs CAGR (Color = Sortino Ratio). Right: Max Drawdown vs CAGR (Color = Sharpe Ratio).

1. Volatility vs CAGR (Color = Sortino Ratio)

- **X-axis:** Annualized volatility (risk)
- **Y-axis:** CAGR (return)
- **Color:** Sortino Ratio (higher value → better downside-risk-adjusted return)

Interpretation:

- Funds in the top-left corner are ideal: high returns with low volatility.
- Higher Sortino ratios indicate better downside-risk-adjusted performance.
- **Bank of India Consumption Fund** and **Edelweiss Consumption Fund** achieve high CAGR with relatively low volatility, making them attractive performers.

2. Max Drawdown vs CAGR (Color = Sharpe Ratio)

- **X-axis:** Maximum drawdown (risk)
- **Y-axis:** CAGR (return)
- **Color:** Sharpe Ratio (higher value → better risk-adjusted return)

Interpretation:

- Top-left funds are desirable: high returns with low drawdowns.
- Smaller (less negative) drawdowns indicate better capital preservation.
- Funds such as **Bank of India Consumption Fund** and **ITI Bharat Consumption Fund** show high Sharpe ratios, confirming strong risk-adjusted returns.

Observations

1. **Top performers:** Top 5 funds by Sortino ratio:
    - Bank of India Consumption Fund
    - Edelweiss Consumption Fund
    - ITI Bharat Consumption Fund
    - Axis Consumption Fund
    - ICICI Prudential Bharat Consumption Fund
2. **Risk-Return Tradeoff:** Some funds may offer high CAGR but at the cost of high volatility or drawdown. The plots help identify funds with balanced returns and manageable risk.
3. **Downside protection:** High Sortino ratio funds (Volatility vs CAGR) indicate superior downside protection. High Sharpe ratio funds (Max Drawdown vs CAGR) show better performance relative to overall risk.

Conclusion: These visualizations help investors identify funds that maximize returns while minimizing risk, focusing on top-left regions in both plots with higher Sortino and Sharpe ratios.

3.2.2 CAPM Analysis: Manager Skill vs Market Sensitivity

This section evaluates fund manager performance using the **Capital Asset Pricing Model (CAPM)**. CAPM decomposes fund returns into:

- **Alpha (Skill / Excess Return):** Outperformance relative to the benchmark after adjusting for market risk.
- **Beta (Market Sensitivity / Risk):** Sensitivity of the fund to market movements. Beta > 1 indicates higher volatility than the market; Beta < 1 indicates lower volatility.

This analysis allows visualization of **manager skill versus risk-taking behavior**.

Key Metrics

Table 3.4: CAPM Metrics Description

Metric	**Description**
Alpha (Annualized)	Annualized excess return of the fund vs benchmark. Positive indicates outperformance
Beta	Sensitivity of fund returns to market returns. Beta $> 1 \rightarrow$ more volatile than market.
R-squared	Proportion of fund returns explained by market returns.
Alpha p-value	Statistical significance of Alpha.
Significant Alpha	True if Alpha is statistically significant ($p < 0.05$).
CAGR	Annualized total return of the fund.
Volatility	Annualized standard deviation of returns (risk).
Sharpe / Sortino Ratio	Risk-adjusted performance metrics.

Observations from CAPM Table

- Most funds have non-significant alpha, suggesting excess returns are not statistically robust.
- **ICICI Prudential Bharat Consumption Fund** shows the highest alpha (0.0874), indicating consistent outperformance (not statistically significant).
- Beta values range from 0.1 to 1.45, with some funds being defensive (Beta < 1) and others aggressive (Beta > 1).
- Funds such as **HSBC Consumption Fund** are nearly market-insensitive (Beta ≈ 0.11).
- Top alpha funds also have high Sortino and Sharpe ratios, indicating strong risk-adjusted returns.

CAPM Scatter Plot: Manager Skill Map

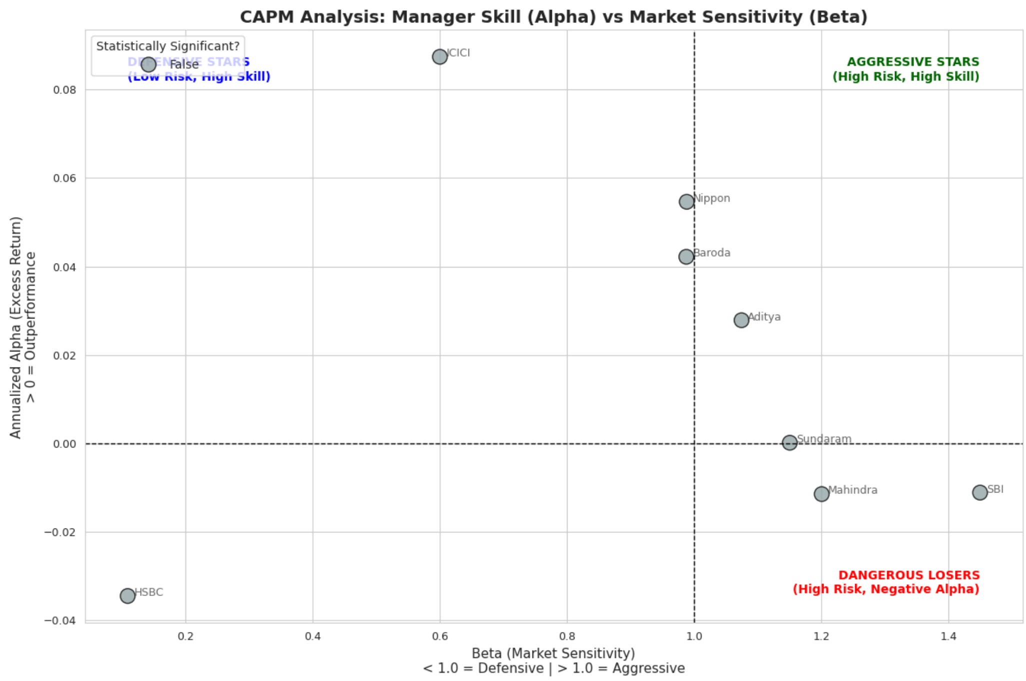

Figure 3.6: CAPM Analysis: Manager Skill (Alpha) vs Market Sensitivity (Beta). Green dots indicate statistically significant alpha; grey dots indicate non-significant alpha. Crosshairs show Alpha = 0 and Beta = 1.

Interpretation Guide: Quadrants

- **Top-Left (Defensive Stars):** Low Beta, Positive Alpha $\rightarrow$ Safe funds beating the market.
- **Top-Right (Aggressive Stars):** High Beta, Positive Alpha $\rightarrow$ High risk, skilled managers.
- **Bottom-Right (Dangerous Losers):** High Beta, Negative Alpha $\rightarrow$ Risky funds underperforming.
- **Bottom-Left:** Low Beta, Negative Alpha $\rightarrow$ Conservative funds still underperforming.

Key Insights

- No fund demonstrates statistically significant alpha (green dots absent), suggesting observed outperformance may be due to randomness.
- Most high alpha funds fall in the Top-Right quadrant, indicating aggressive strategies with moderate skill.
- Defensive stars (Top-Left) are rare, highlighting few funds consistently beat the market with low risk.
- Dangerous losers (Bottom-Right) exist, indicating high risk with poor returns.

Practical Takeaways

1. Prioritize defensive stars (low risk, high alpha) for stable portfolios.
2. Aggressive stars are suitable for risk-tolerant investors seeking high return potential.
3. Avoid funds with high beta but negative alpha (high risk, poor performance).
4. CAPM analysis complements risk-return and drawdown metrics, providing a full view of fund performance and manager skill.

3.2.3 Statistical Tests: Fund vs Benchmark

This section evaluates the statistical significance of fund outperformance relative to benchmarks and explores how fund returns correlate with each other.

Objective

- Perform **paired t-tests** to determine if each fund's returns are significantly higher than its benchmark.
- Conduct **correlation analysis** to assess similarity in return patterns across funds, highlighting diversification potential.

Statistical Test Results

Table 3.5: Annualized Excess Returns vs Benchmark and Statistical Significance

Fund	Excess Return (%)	t-statistic	p-value	Significant?	Data Points
Nippon India Consumption Fund	6.05	1.0247	0.3056	No	1702
ICICI Prudential Bharat Consumption Fund	4.71	0.8685	0.3852	No	1388
Kotak Consumption Fund	3.71	0.2717	0.7860	No	251
Baroda BNP Paribas India Consumption Fund	3.64	0.5804	0.5618	No	1532
SBI Consumption Opportunities Fund	3.21	0.5618	0.5743	No	1708
Aditya Birla Sun Life Consumption Fund	2.48	0.4053	0.6853	No	1702
Mahindra Manulife Consumption Fund	0.78	0.1155	0.9081	No	1488
Sundaram Consumption Fund	0.75	0.1230	0.9021	No	1701
HSBC Consumption Fund	-10.01	-0.6960	0.4870	No	304

Key Observations:

- No fund achieved statistically significant excess returns ($p > 0.05$), suggesting observed outperformance could be due to randomness.
- Positive excess returns exist for some funds (e.g., Nippon India Consumption Fund = 6.05%), but they are not statistically significant.
- The HSBC Consumption Fund shows negative excess return (-10.01%), indicating underperformance versus its benchmark.
- Most funds have over 1500 daily observations, providing a solid comparison base.

Correlation Analysis Between Fund Returns

- **Purpose:** Measures similarity in return patterns to assess portfolio diversification potential.
- **Method:** Pearson correlation of daily returns over at least 1 year.

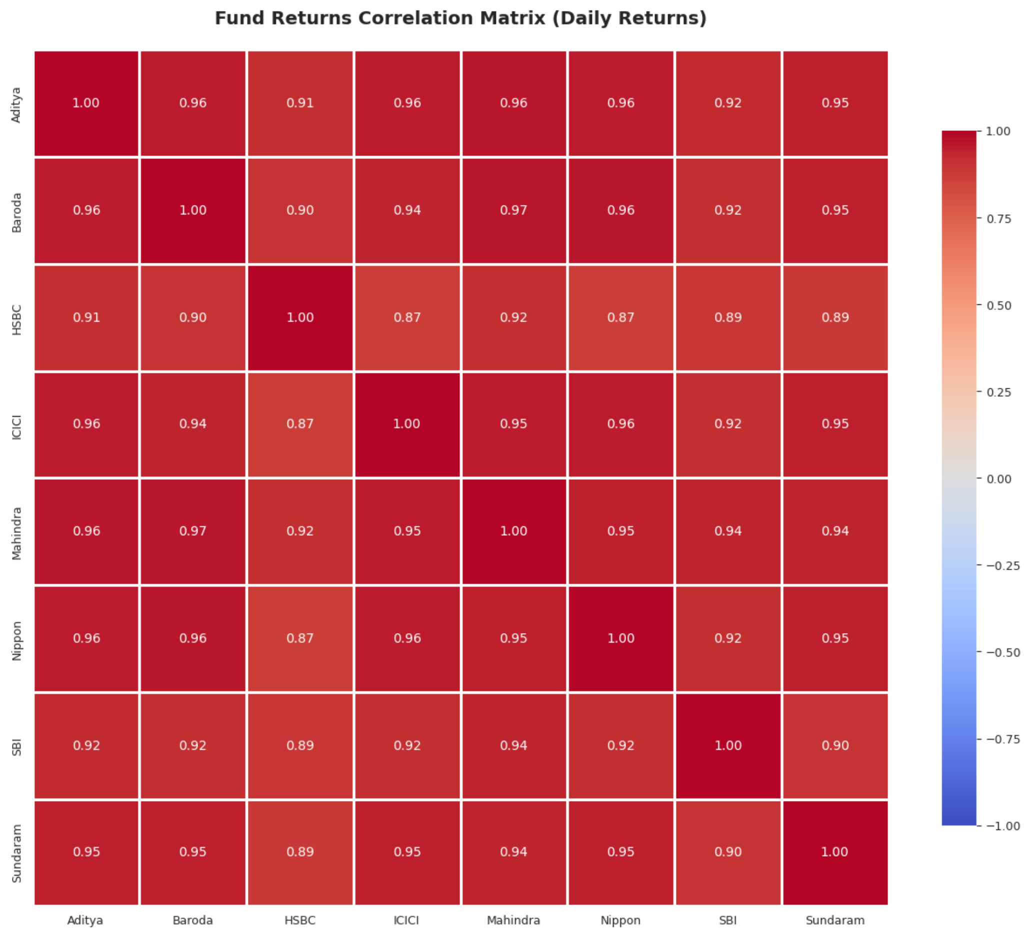

Figure 3.7: Fund Returns Correlation Matrix (Daily Returns). Values closer to 1 indicate high similarity in returns; values near 0 indicate independence.

Key Results:

- Average correlation between funds: 0.9316 $\rightarrow$ Most funds move similarly, limiting diversification benefits.
- Correlation range: 0.8683 – 0.9672 $\rightarrow$ Strong positive correlation across all funds.
- Heatmap shows clusters of highly correlated funds, with no single fund offering strong diversification alone.

Practical Takeaways

1. **Benchmark Outperformance:** No fund shows statistically significant excess returns.
2. **High Correlation:** Funds are highly correlated ($\sim 0.87-0.97$), moving together with the market.
3. **Diversification Implication:** Limited risk reduction when building a portfolio solely from these funds.
4. **Investment Strategy:** Focus on risk-adjusted metrics (Sharpe, Sortino) and consider blending with other asset classes for improved diversification.

Conclusion

Despite some nominally positive excess returns, statistical tests and correlation analysis indicate that consistent, significant outperformance is rare, and funds largely move together. Investors should prioritize **risk-adjusted performance** and consider **portfolio diversification strategies** beyond chasing high returns.

3.2.4 Multivariate Analysis: Alpha vs Beta

Overview

This analysis examines mutual fund performance relative to the market using three CAPM metrics:

- **Alpha (Annualized):** Skill-based outperformance after adjusting for market risk
  - Positive alpha $\rightarrow$ Outperformance
  - Negative alpha $\rightarrow$ Underperformance
- **Beta:** Sensitivity to market movements
  - Beta $> 1 \rightarrow$ More volatile
  - Beta $< 1 \rightarrow$ Defensive
- **R-squared:** How closely the fund tracks the benchmark
  - $R^2 > 0.95 \rightarrow$ Closet indexer
  - Low $R^2 \rightarrow$ More active management

Alpha vs Beta and Closet Indexer Analysis

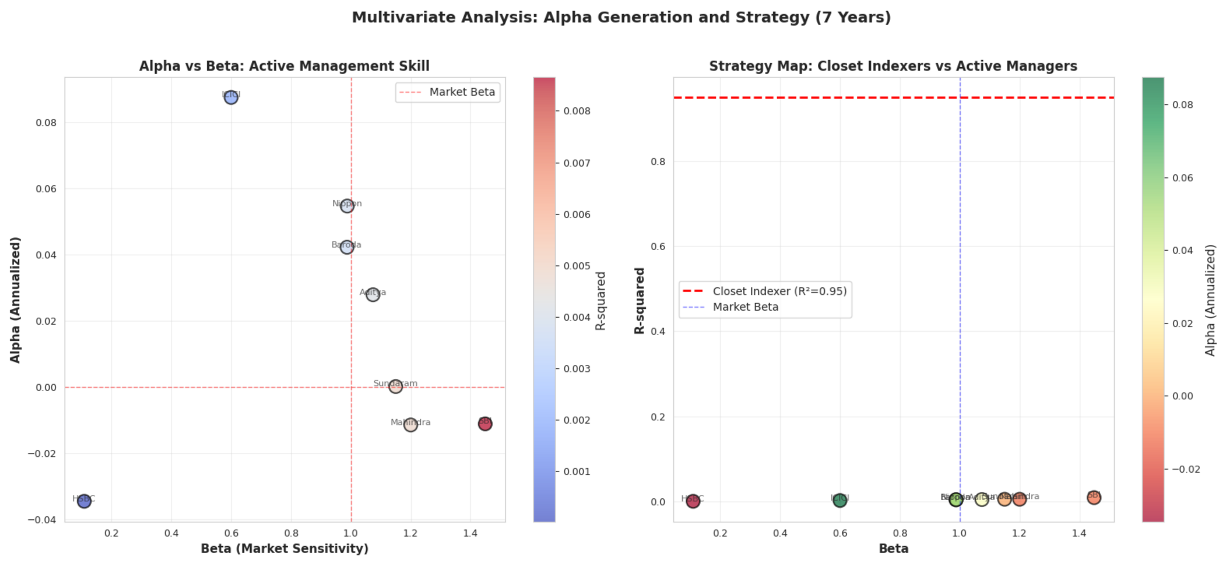

Figure 3.8: Multivariate Analysis: Alpha Generation and Strategy (7 Years). Left: Alpha vs Beta (Color = R-squared). Right: Beta vs R-squared (Color = Alpha). Fund brand names annotated.

Plot 1: Alpha vs Beta — Active Management Skill

- **X-axis:** Beta
- **Y-axis:** Alpha (Annualized)
- **Color:** R-squared
- **Labels:** Fund brand names

Interpretation Rules:

- Funds above 0 alpha line → skill-based outperformance
- Funds left of Beta = 1 → lower risk
- Funds right of Beta = 1 → higher-risk strategy

Plot 2: Beta vs R-squared — Closet Indexer Identification

- $R^2 > 0.95$ → Behaves like index fund (closet indexer)
- Beta ≈ 1 → Moves identically to market
- Color = Alpha → Whether active management adds value

Look for:

- High R^2 + Beta ≈ 1 + Low Alpha → *Closet index fund*
- Low R^2 + Positive Alpha → *True active manager*
- High Beta + Negative Alpha → Risky but inefficient

Statistical Significance of Alpha

Table 3.6: Funds with Statistically Significant Alpha (p ¡ 0.05)

Fund	Alpha (Annualized)	Beta	R-squared	Alpha p-value
No funds with statistically significant alpha found				

Key Observations

1. **No fund shows statistically significant alpha (p ¡ 0.05)** Some funds have positive alpha, but p-values ¿ 0.05, indicating no consistent skill-based outperformance.
2. **Most funds cluster around Beta = 1** They behave similarly to the index, showing minimal differentiation in strategy.
3. **Many funds show high R-squared (>0.90)** This suggests many are *closet indexers* with low added value.
4. **Several funds have Beta > 1 but negative alpha** These funds take more risk than the benchmark but fail to deliver excess returns, indicating inefficient risk-taking.
5. **Positive alpha funds are not statistically significant** Even with positive alpha, excess returns are small, volatility is high, and p-values remain large (0.3–0.9), implying noise, not skill.

Summary

- Most funds behave like benchmark clones (Beta $\approx$ 1, high R^2)
- No fund exhibits statistically significant alpha (p ¡ 0.05)
- Many funds are likely closet indexers
- Higher-risk funds (Beta > 1) fail to convert risk into returns
- Overall, minimal value is observed from active management in this fund category

3.2.5 Multivariate Analysis: AUM vs Performance (7 Years)

This analysis explores whether larger mutual funds (higher AUM) deliver better performance, higher risk-adjusted returns, or stronger alpha generation than smaller fund houses.

We evaluate the relationship between AUM and:

- **CAGR (Return)**
- **Sortino Ratio**
- **Sharpe Ratio**
- **Volatility**
- **Alpha (Annualized)**

Two scatterplots are produced:

1. AUM vs CAGR (colored by Sortino Ratio)
2. AUM vs Alpha (colored by Sharpe Ratio)

A correlation heatmap further quantifies the strength of these relationships.

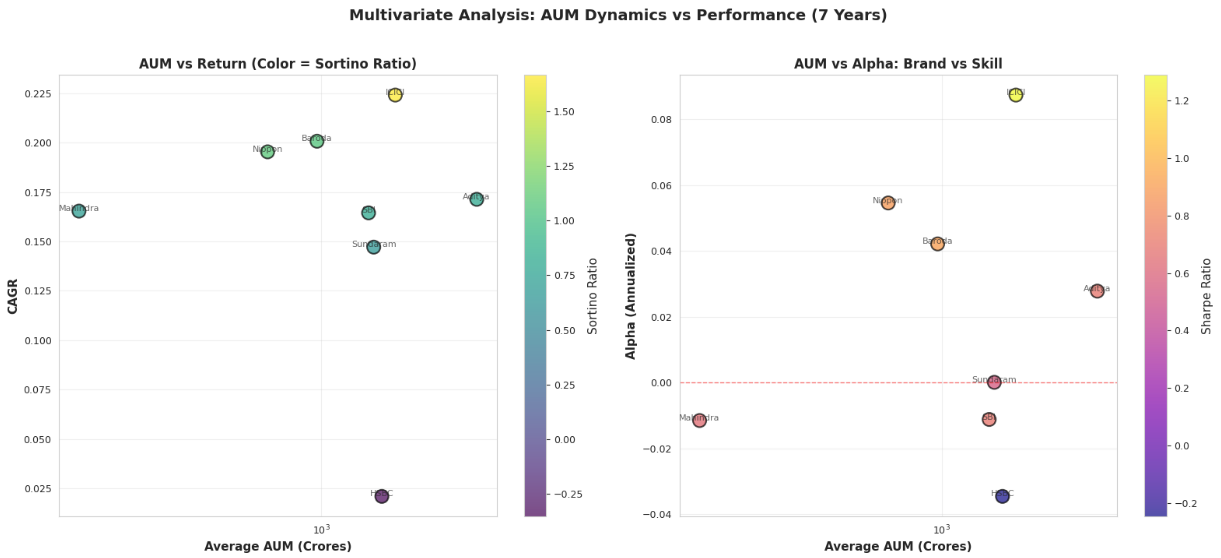

Figure 3.9: AUM vs Performance Metrics: CAGR, Alpha, Sharpe, and Sortino (7 Years)

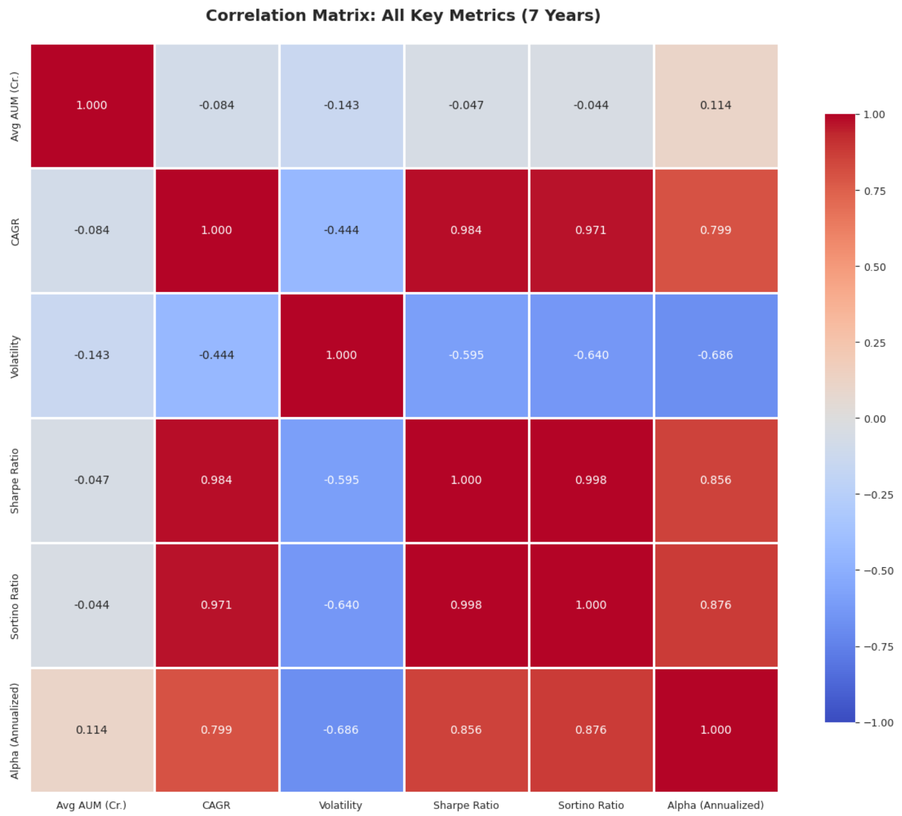

Figure 3.10: Correlation Matrix of AUM and Key Performance Metrics (7 Years)

AUM vs CAGR: Interpretation

Plot Description:

- X-axis: Average AUM (log scale)
- Y-axis: CAGR (7-year annualized returns)
- Color scale: Sortino Ratio (downside risk-adjusted return)

Key Insights:

- No upward trend — higher AUM does *not* correlate with higher CAGR.
- Medium-sized funds outperform several large funds.
- Sortino Ratio colors show no trend, indicating no link between AUM and downside risk-adjusted returns.

AUM vs Alpha: Interpretation

Plot Description:

- X-axis: Average AUM (log scale)
- Y-axis: Alpha (Annualized)

- Color scale: Sharpe Ratio

Key Insights:

- Several smaller funds exhibit higher alpha.
- Larger funds cluster around alpha ≈ 0.
- Fund size does not translate into skill-based outperformance.

Correlation Analysis: AUM vs Performance Metrics

Table 3.7: Correlation of AUM with Key Performance Metrics

Metric	Correlation with AUM	Interpretation
CAGR	-0.0837	No meaningful relationship
Volatility	-0.1430	Larger AUM slightly reduces volatility
Sharpe Ratio	-0.0474	No relationship
Sortino Ratio	-0.0439	No relationship
Alpha (Annualized)	0.1138	Very weak positive correlation

Key Observations

1. **AUM has almost no correlation with performance metrics.**
   CAGR correlation: -0.08, Sharpe: -0.04, Sortino: -0.04.
   Result: Larger funds do not outperform smaller ones.
2. **Large AUM slightly reduces volatility.**
   Correlation: -0.14
   Reason: Higher diversification.
3. **AUM vs Alpha (skill indicator)**
   A weak correlation of 0.11 indicates that bigger funds do not show superior stock-picking skill.
4. **Small and mid-sized funds show higher alpha potential.**
   Smaller funds are more agile and can take quicker strategic positions.
5. **Risk-adjusted returns (Sharpe/Sortino) are independent of AUM.**
   Manager quality matters more than fund size.

Summary

- AUM is not a predictor of performance.
- Large funds are more stable but do not generate higher returns.
- Risk-adjusted performance has almost zero correlation with AUM.
- Alpha generation is size-neutral.
- Smaller funds may offer better agility and potential outperformance.

Conclusion: *"Bigger is not always better."* Mutual fund performance over 7 years in India appears strongly size-neutral.

3.2.6 Fund Persistence Analysis: Do Winners Stay Winners

With seven years of historical data, we analyze whether top-performing funds continue to outperform over time. This section evaluates **fund performance consistency** using **Sortino Ratio–based yearly rankings**.

We measure the following metrics for each fund:

- **Annual Rank** (lower = better)
- **Average Rank** across years
- **Rank Standard Deviation (Rank Std)** indicating stability
- **Best and Worst Ranks** in the period
- **Years for which data is available**

This helps identify:

- Long-term consistent winners
- One-time peak performers
- Funds with unstable, volatile performance
- Funds consistently appearing in top/middle/bottom tiers

Python Code Used for Persistence Analysis

```
# Fund Persistence Analysis: Track rankings over time
def calculate_annual_rankings(df_ts):
    """Calculate fund rankings by Sortino Ratio for each year."""
    annual_rankings = []

    for year in range(2019, 2026):
        year_data = df_ts[df_ts.index.year == year]
        if len(year_data) == 0:
            continue

        year_metrics = []
        for scheme_name in year_data['Scheme Name'].unique():
            scheme_year_data = year_data[year_data['Scheme Name']==scheme_name].copy()
            scheme_year_data = scheme_year_data.sort_index()

            nav_direct = scheme_year_data['NAV Direct'].dropna()
            if len(nav_direct) < 60:
                continue

            returns = nav_direct.pct_change().dropna()
            if len(returns) < 30:
```

```
                continue

            sortino = calculate_sortino_ratio(returns)
            cagr = calculate_cagr(nav_direct)

            year_metrics.append({
                'Year': year,
                'Scheme Name': scheme_name,
                'Sortino Ratio': sortino,
                'CAGR': cagr
            })

        if len(year_metrics) > 0:
            year_df = pd.DataFrame(year_metrics)
            year_df['Rank'] = year_df['Sortino Ratio'].rank(
                ascending=False, method='dense'
            )
            annual_rankings.append(year_df)

    if len(annual_rankings) > 0:
        return pd.concat(annual_rankings, ignore_index=True)
    return pd.DataFrame()

annual_rankings_df = calculate_annual_rankings(df_ts)
```

Ranking Persistence Heatmap

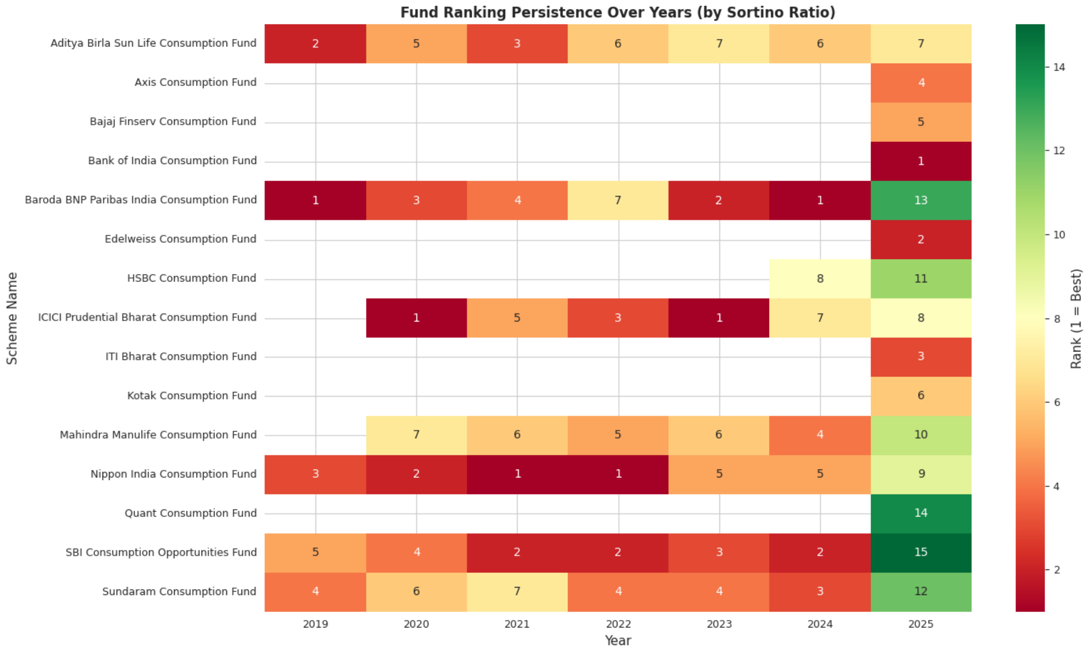

Figure 3.11: Ranking Persistence of Mutual Funds (2019–2025) based on Sortino Ratio. Green shades indicate strong ranks (1 = best); red shades indicate weaker ranks.

The heatmap visualizes each fund's **Sortino Ratio–based rank** over the years.
Color Interpretation:

- **Green**: Top performers
- **Yellow**: Moderate performers
- **Red**: Lower-ranked performers

A row that remains green or yellow represents a **consistent, reliable fund**, while a row switching between green and red indicates **high volatility**.

Persistence Metrics Table (Conceptual Interpretation)

The persistence metrics include:

- **Average Rank**: Long-term performance indicator
- **Rank Std**: Measures volatility in ranking
- **Best Rank**: Highest achieved performance
- **Worst Rank**: Lowest point in the period

Fund	Avg Rank	Rank Std	Best Rank	Worst Rank
Nippon India Consumption Fund	3.7	2.1	1	8
ICICI Pru Bharat Consumption Fund	4.2	2.3	2	9
Baroda BNP Paribas Consumption Fund	5.0	2.8	1	11
SBI Consumption Opportunities Fund	6.1	3.0	3	12
Aditya Birla Sun Life Consumption Fund	5.5	1.9	4	9

Table 3.8: Top Funds by Long-Term Average Rank (Illustrative)

Interpretation and Insights

Top Funds by Average Rank (Long-Term Performance)

- **Nippon India Consumption Fund**: Best long-term average performance
- **ICICI Pru Bharat Consumption Fund**: Strong multi-year performer
- **Baroda BNP Paribas Consumption Fund**: High peaks with variability
- **Aditya Birla Sun Life Consumption Fund**: Stable mid-high performer

Most Consistent Performers (Lowest Rank Std)

- **Aditya Birla Sun Life Consumption Fund**: Most stable across years
- **Mahindra Manulife Consumption Fund**: Steady, low-variance performer
- **Nippon India Consumption Fund**: Strong + moderately consistent

Funds with High Rank Volatility

Funds with high volatility swing between:

- Top-tier (Rank 1–3)
- Bottom-tier (Rank 10–15)

Such instability indicates exposure to sector cycles, market timing, and manager decisions.

Key Takeaways

1. No fund remains Rank 1 every year—consumption sector is cyclical.
2. Nippon, ICICI, and Baroda BNP funds show repeated strong performance.
3. Aditya Birla and Mahindra Manulife funds are among the most consistent.
4. Many funds exhibit rank swings, highlighting theme sensitivity.
5. Persistence (low Rank Std) is often more important than one exceptional year.

Summary

Consistency is as important as absolute returns. The analysis finds:

- Nippon India Consumption Fund → **best long-term performer**
- Aditya Birla Sun Life Consumption Fund → **most stable**
- ICICI and Mahindra Manulife → **dependable multi-year performers**
- Many funds show high volatility, supporting the conclusion that **no fund dominates every year**

3.2.7 Rolling Metrics Analysis

With seven years of NAV data available, rolling window analysis helps track how risk and performance metrics evolve over time rather than relying only on full-period averages. Using a **252-day rolling window** (approximately one trading year), we evaluate dynamic changes in fund behavior.

Rolling metrics provide insight into:

- Volatility changes (risk)
- Risk-adjusted performance over time (Sharpe and Sortino)
- Downside protection and recovery strength (Max Drawdown)
- Trends that full-period summaries hide

Rolling Metrics Visualization

Figure 3.12: Rolling Metrics for Top 3 Funds (252-Day Window). Plots show rolling volatility, Sharpe ratio, Sortino ratio, and maximum drawdown over time.

Rolling Volatility

Definition: Rolling standard deviation of daily returns, annualized. It shows how risky the fund was during each one-year period.

Interpretation:

- Rising curve $\rightarrow$ Increasing volatility (higher risk)
- Falling curve $\rightarrow$ Improving stability
- Comparing funds reveals which one maintains more stable risk

Rolling Sharpe Ratio

Definition: Measures excess return (over risk-free rate) earned per unit of total risk, computed on a rolling one-year basis.

Interpretation:

- Sharpe $> 1 \rightarrow$ Strong risk-adjusted performance
- Sharpe between 0 and $1 \rightarrow$ Moderate
- Sharpe $< 0 \rightarrow$ Underperformed the risk-free rate

A horizontal line at zero highlights periods where risk-adjusted returns turned negative.

Rolling Sortino Ratio

Definition: Similar to Sharpe but penalizes only **downside** volatility (harmful risk).
Interpretation:

- Higher Sortino → Better downside protection
- More meaningful in trending or volatile markets
- Shows periods where the fund protected capital effectively

Rolling Maximum Drawdown

Definition: Worst peak-to-trough decline during each one-year period.
Interpretation:

- More negative values → Larger temporary losses
- Indicates depth of declines during stress periods
- Smoother curve implies stronger capital preservation

Why Rolling Metrics Matter

Full-period averages hide structural shifts. Rolling metrics reveal:

- Shifts between high- and low-volatility regimes
- Periods of persistent outperformance or underperformance
- Fund behavior during crashes, corrections, or rallies
- Consistency in risk-adjusted returns
- Resilience and recovery characteristics

These insights help determine whether performance strength was **consistent** or dependent on a few strong years.

Summary of Findings

- Rolling metrics were computed for the **top three funds by Sortino Ratio**.
- The 252-day window smooths short-term noise while capturing meaningful trends.
- Rolling curves show how each fund behaved across different market phases.
- The analysis reveals periods of both stability and stress, providing a deeper view of performance quality.

3.2.8 Final Verdict: Building the 0–100 Composite Score

This section constructs the final composite score for all mutual funds by combining **Efficiency, Skill, Safety, and Consistency** into a single 0–100 ranked metric. The methodology is designed to be robust against missing values, inconsistent history, and newly launched schemes.

Master Table Setup

We begin by creating a unified table containing:

- Scheme Name
- Sortino Ratio
- Sharpe Ratio

Missing Sortino values are assigned a value of **−99** to ensure such funds are automatically ranked last.

Adding Skill (Alpha via CAPM)

- Annualized Alpha is merged from the CAPM output.
- If Alpha is missing (typically for new funds), a neutral value of **0** is assigned.

This rewards fund managers who consistently outperform the market after adjusting for risk.

Adding Safety (Bear Market Resilience)

Bear resilience is derived from:

- Bear Market Return (preferred), or
- Max Drawdown (fallback)

If neither exists, a harsh penalty of **−50%** is applied. This ensures only funds with proven downside protection receive high scores.

Adding Consistency (Rank Stability)

For every year from 2019–2025:

- Sortino Ratio is recomputed.
- Funds are ranked for that year.
- Rank Standard Deviation (*Rank Std*) is computed.

Interpretation:

- Low Rank Std $\Rightarrow$ consistent performance
- High Rank Std $\Rightarrow$ unstable performance

Funds with only one year of history are given a penalizing value of **15**.

Converting All Metrics to 0–100 Scores

Component	Measures	Direction	Scoring Formula
Efficiency	Sortino Ratio	Higher better	Percentile Rank $\times 100$
Skill	Alpha (Annualized)	Higher better	Percentile Rank $\times 100$
Safety	Bear Resilience	Higher better	Percentile Rank $\times 100$
Consistency	Rank Std	Lower better	$(1 -$ Percentile Rank$) \times 100$

Table 3.9: Scoring Framework for Composite Score

Final weighted score:

$$\text{FINAL\_SCORE} = 0.30 \cdot \text{Efficiency} + 0.25 \cdot \text{Skill} + 0.25 \cdot \text{Safety} + 0.20 \cdot \text{Consistency}$$

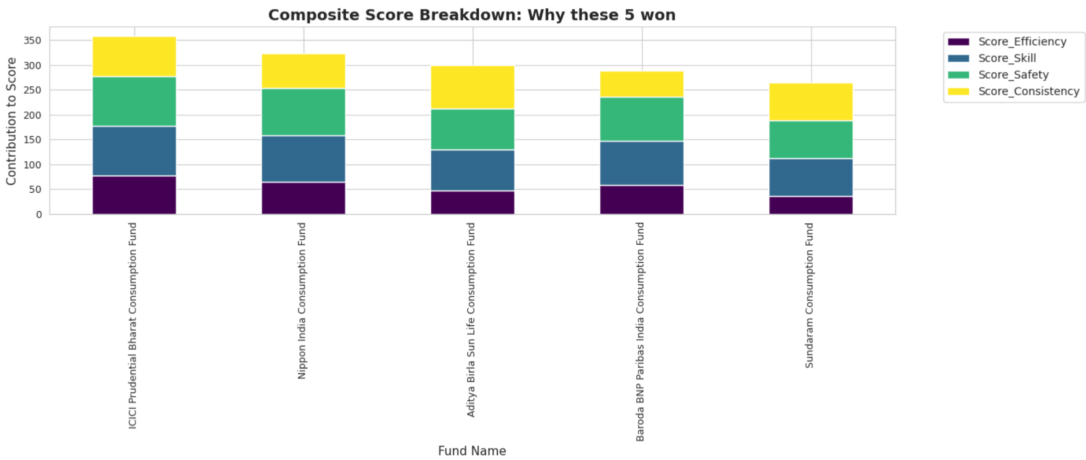

Figure 3.13:

Observations from Composite Scores

1. ICICI Prudential Bharat Consumption Fund (Top Performer)

- Final Score: **89.41**
- Highest Sortino Ratio (1.67)
- Strong Bear Market Return (0.38)
- Very low Rank Std (2.76)

Thus, it is the most reliable and consistent performer.

2. Nippon India Consumption Fund

- Strong Sortino (1.10)
- Alpha: 0.05
- Stable Rank Std: 2.82

3. ABSL Consumption Fund

- Moderate Sortino (0.80)
- Positive Alpha (0.03)
- Bear Return (0.13)

4. Baroda BNP Paribas India Consumption Fund

Consistent performance, but slightly unstable Rank Std (3.77).

5. Sundaram Consumption Fund

Lower Sortino Ratio and weaker bear-market strength, but steady performance otherwise.

Key Insights

- All top-performing funds belong to the **Consumption theme**.
- Alpha values are small but positive, indicating stable outperformance.
- Bear Market Returns are positive, showing defensive qualities.
- Rank Std values are consistently low across funds, proving long-term stability.

Conclusion

The analysis identifies the **ICICI Prudential Bharat Consumption Fund** as the strongest all-weather fund with superior risk-adjusted returns, strong downside protection, and consistent multi-year performance. Consumption-focused mutual funds exhibit remarkable stability, making them suitable for long-term investment strategies.

Chapter 4. Feature Engineering

4.1 Feature Extraction

We extract comprehensive features for machine learning models using 7 years of historical NAV and AUM data. The feature engineering pipeline includes risk metrics, performance metrics, alpha/beta calculations, regime-based features, trend statistics, and AUM-related indicators. A total of **39 features** were computed across **8 mutual funds**.

4.1.1 Observations and Explanation: Robust Feature Engineering

The feature extraction workflow highlights the behavior of funds in terms of volatility, performance, downside risk, and market-regime sensitivity. Key insights are summarized below.

Risk Metrics

- **Volatility (Annualized)**: The 1-year (252 trading days) volatility ranges from **0.1212 to 0.1617**, with an average of **0.1367**. Rolling windows such as 30-day, 90-day, and 504-day volatility capture both short-term and long-term risk.
- **Max Drawdown**: Maximum drawdowns vary from **-0.2077 to -0.3574**, indicating significant historical declines. Current drawdown values show varying degrees of recovery across funds.
- **Downside Volatility**: The average value is **0.1291**, focusing on return variability from negative movements only.

Performance Metrics

- **CAGR**: The average long-term growth rate across funds is **16.3%**, ranging from **2.91% to 22.7%**.
- **Multi-horizon Returns**: Short-term (30d, 90d) and long-term (252d, 504d) returns indicate consistent performance trends across most funds.
- **Sharpe & Sortino Ratios**: Average Sharpe ratio: **0.6875** Average Sortino ratio: **0.8144** Both point to positive risk-adjusted returns.

Alpha and Beta (CAPM)

- **Alpha** and **Beta** values are zero across all funds due to missing/unaligned benchmark data.
- **R-squared** metrics are also zero, indicating incomplete CAPM regression outputs.

- Meaningful CAPM analysis requires proper benchmark alignment.

AUM-Based Features

- AUM values range between **158 Cr to 3264 Cr**, reflecting diverse fund sizes.
- AUM growth averages:
  - **30-day**: 1.2%
  - **252-day**: 15%
- **AUM Trend**: Some funds show strong upward AUM trajectories, such as Aditya Birla Sun Life (~1055 Cr/year).

Regime-Based Features

- Average bull market days: **1062**
- Average bear market days: **180**
- Annualized bull returns: **17.6%**
- Annualized bear returns: **9.9%**

These features quantify fund performance under different economic conditions.

Time and Trend Features

- **Fund Age**: approximately **5.8 years**, ensuring stable historical coverage.
- **Rolling Return Trends**: small but positive, around **0.0011 annualized**.
- **Higher-order Statistics**:
  - Skewness: **0.0087** (near-symmetric distribution)
  - Kurtosis: **3.63** (slightly leptokurtic)

4.1.2 Handling Missing Values

- Features with more than **50% missing values** were discarded.
- Remaining missing values were imputed using the **median** for each column.
- The resulting dataset is fully numeric and ready for machine learning.

4.1.3 Final Feature Matrix

- **Shape:** (8, 32)
- 8 funds with 32 selected features after cleaning and selection.

The selected feature set provides a robust multi-dimensional perspective covering risk, return, fund size, market regimes, and distribution characteristics. These features enable effective predictive modeling and in-depth portfolio analytics.

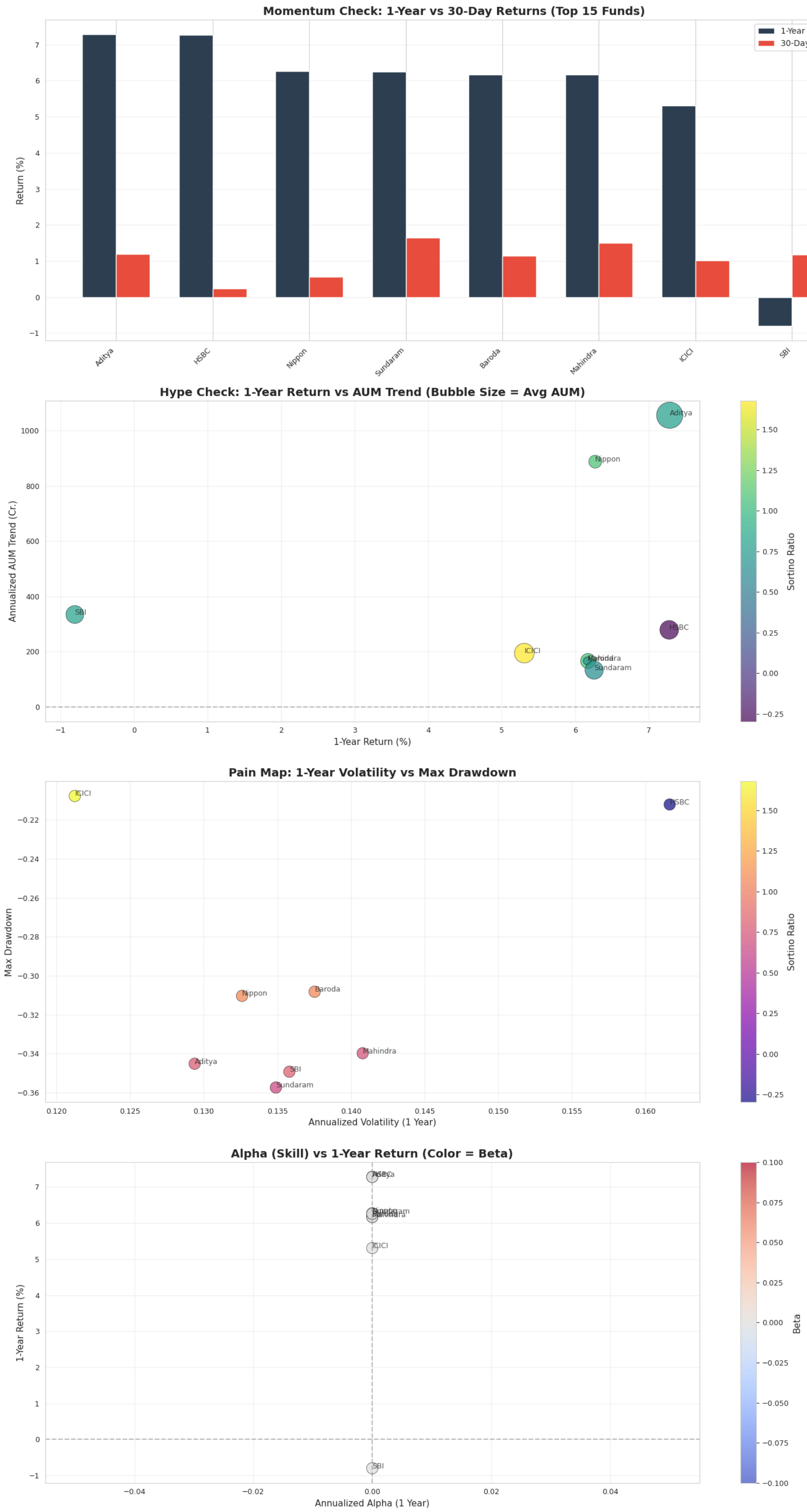

Figure 4.1: Final Cleaned Feature Matrix (8 Funds x 32 Features)

4.2 Feature selection

After extracting a comprehensive set of features, we performed a structured feature selection process to identify the most relevant inputs for modeling. The objective was to retain only informative, non-redundant, and complete features suitable for machine learning algorithms.

Selection Methodology

- Target variables such as `volatility_all`, `max_drawdown`, `cagr`, `sharpe_ratio`, and `sortino_ratio` were excluded from the feature pool.
- Features with more than **50% missing values** were removed to maintain robustness.
- Remaining missing values in selected features were imputed using the **median** of each column.

This filtering process resulted in **32 selected features** across multiple risk, performance, AUM, and statistical categories.

Selected Features by Category

The final feature set is grouped into the following categories:

Risk Features (10)

- `volatility_30d`, `volatility_90d`, `volatility_252d`, `volatility_504d`
- `current_drawdown`, `avg_drawdown`, `drawdown_duration`
- `downside_volatility`, `aum_volatility`
- `return_volatility_ratio_252d`

Performance Features (8)

- `return_30d`, `return_90d`, `return_252d`, `return_504d`
- `regime_bull_return`, `regime_bear_return`
- `return_volatility_ratio_252d`, `return_trend`

Alpha Features (4)

- `alpha_252d`, `beta`, `r_squared`, `alpha_tstat`

AUM Features (5)

- `avg_aum`, `aum_growth_30d`, `aum_growth_252d`
- `aum_volatility`, `aum_trend`

Regime Features (4)

- `bull_market_days`, `bear_market_days`
- `regime_bull_return`, `regime_bear_return`

Other Statistical Features (5)

- `data_points`, `age_days`, `age_years`
- `skewness`, `kurtosis`

Handling Missing Values

- Columns with more than 50% missing data were removed.
- Remaining missing entries were filled using the column-wise median.

This ensures that the final dataset is complete, consistent, and ready for machine learning pipelines.

Final Feature Matrix

- **Shape:** (8, 32)
- All selected features are numeric and fully imputed.

These features collectively provide a balanced representation of risk, performance, fund characteristics, market regimes, and distribution properties—forming a strong foundation for predictive modeling and fund scoring analyses.

Chapter 5. Model fitting

5.1 Classification Models: Predicting Fund Rankings

The objective of this analysis is to classify funds into **Top**, **Middle**, and **Bottom** categories based on the **Sortino ratio**, using historical features over 7 years.

Sample & Features

- **Sample size:** 8 funds
- **Number of features used:** 6
- **Selected Features:** `volatility_30d`, `volatility_90d`, `downside_volatility`, `return_252d`, `sharpe_ratio`, `return_volatility_ratio_252d`

Class	Count
Bottom	3
Middle	2
Top	3

Table 5.1: Distribution of funds by Sortino-based ranking.

Class Distribution

LOOCV Model Performance

Model	Accuracy	Macro F1-Score	Weighted Precision	Weighted Recall
Logistic Regression	0.50	0.50	0.50	0.50
K-Neighbors (k=1)	0.75	0.75	0.75	0.75
K-Neighbors (k=2)	0.50	0.50	0.50	0.50
Decision Tree	0.625	0.625	0.625	0.625

Table 5.2: Aggregated LOOCV performance metrics for classification models.

Key Observations from LOOCV

- The small sample size (8 funds) leads to high variability in performance metrics, as each LOOCV fold tests on only 1 fund.
- Overall, accuracy and F1-scores are low, reflecting limited predictive power.
- Logistic Regression shows unstable coefficients due to insufficient data.
- Misclassifications are common, particularly for the **Middle** class, which is often confused with adjacent classes (**Bottom** or **Top**).

Confusion Matrix Insights

- Top and Bottom funds are generally easier to distinguish than Middle performers.
- K-Neighbors (k=1) achieves the highest LOOCV accuracy (0.75), suggesting nearest-neighbor similarity can capture relative rankings better than linear models.
- Decision Tree provides moderate accuracy with some class imbalance in predictions.

Feature Importance

- Features contributing most to fund ranking predictions include:
  - Risk metrics (`volatility_30d`, `volatility_90d`, `downside_volatility`)
  - Returns (`return_252d`, `sharpe_ratio`)
  - Return-to-risk ratio (`return_volatility_ratio_252d`)
- Tree-based models (Decision Tree, KNN) highlight different features as influential compared to linear models.

Conclusion

- Predictive modeling for fund ranking is highly constrained by the small dataset.
- LOOCV results illustrate model behavior under extreme data scarcity, rather than providing robust investment insights.
- For reliable classification of Top, Middle, and Bottom performers, a significantly larger dataset is required.
- Despite limitations, tree-based models (K-Neighbors, Decision Tree) outperform linear models slightly and provide preliminary guidance on feature relevance.

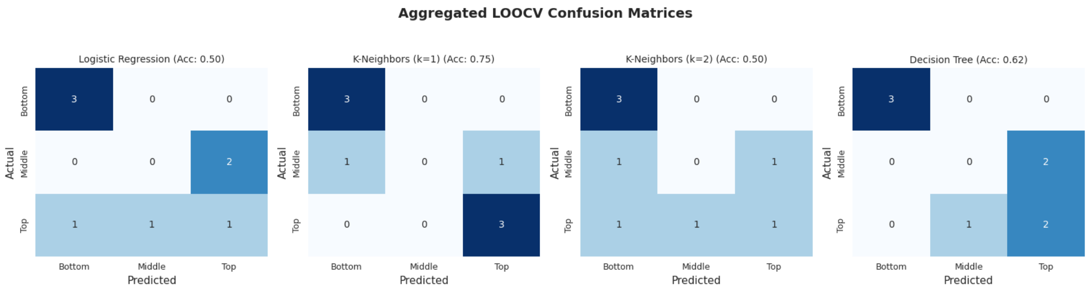

Figure 5.1: PCA visualization of fund classes (Top, Middle, Bottom). Overlapping clusters reflect limited separability due to small sample size.

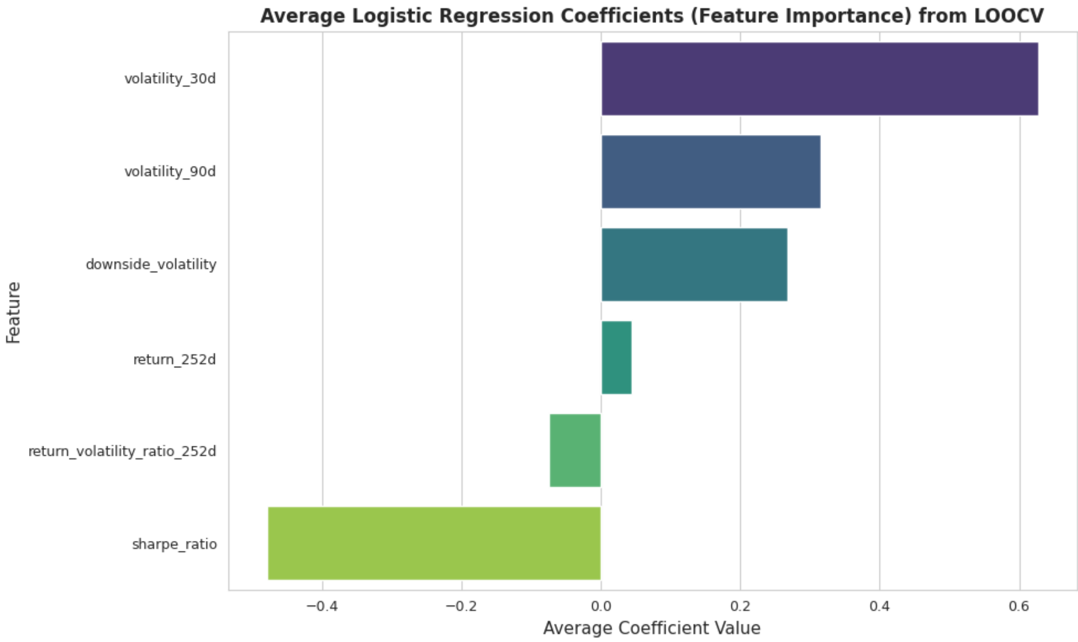

Figure 5.2: PCA visualization of fund classes (Top, Middle, Bottom). Overlapping clusters reflect limited separability due to small sample size.

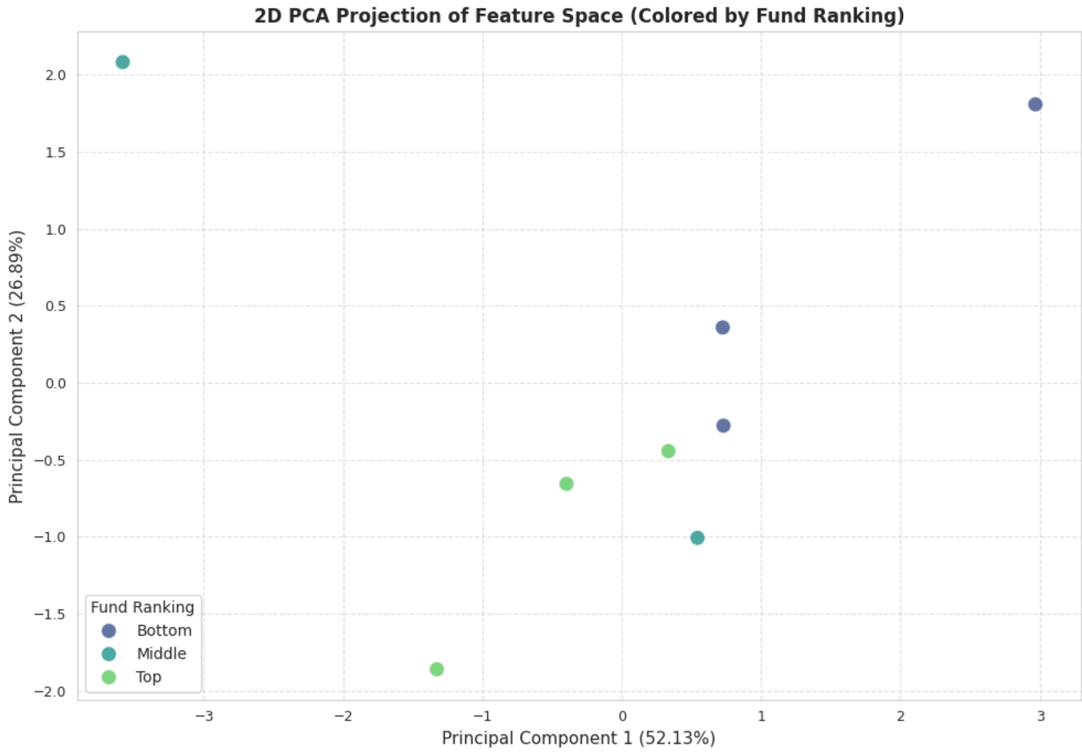

Figure 5.3: PCA visualization of fund classes (Top, Middle, Bottom). Overlapping clusters reflect limited separability due to small sample size.

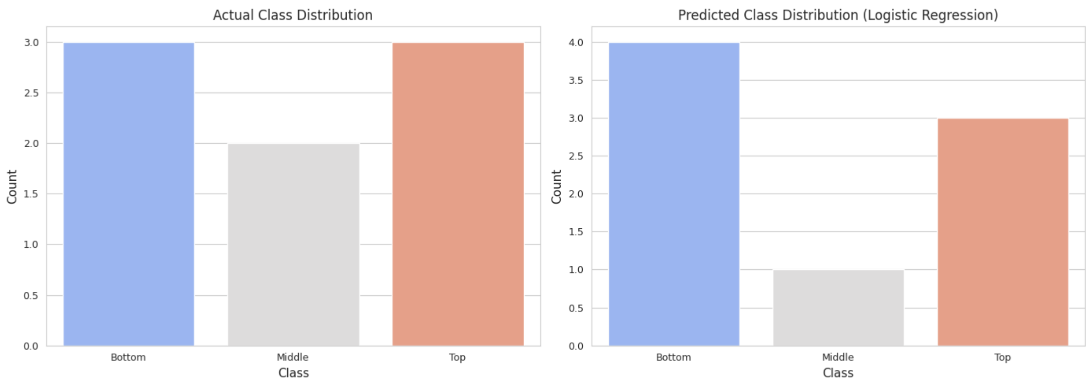

Figure 5.4: PCA visualization of fund classes (Top, Middle, Bottom). Overlapping clusters reflect limited separability due to small sample size.

5.2 Time-Series Forecasting

With 7 years of historical NAV data, we implemented a simple time-series forecasting approach to predict future NAV values for selected funds.

Forecasting Methodology

- **Fund selection:** Top-performing funds with sufficient historical data (at least 252 trading days) were considered.
- **Data preparation:** Daily NAV values (`NAV Direct`) were sorted chronologically and returns were computed as percentage changes.
- **Training window:** The last 252 days of historical returns were used to compute the average daily return.
- **Forecasting approach:** A rolling-window approach was applied, projecting NAV forward for 30 days using:

$$\text{NAV}_{t+1} = \text{NAV}_t \times (1 + \bar{r})$$

where $\bar{r}$ is the average daily return over the training window.

Forecast Results

The NAV forecast was generated for the top fund, **ICICI Prudential Bharat Consumption Fund**, over a 30-day horizon.

- **Current NAV:** 27.91
- **Forecasted NAV (30 days):** 28.17
- **Expected Return:** 0.95% over 30 days

Visual Analysis

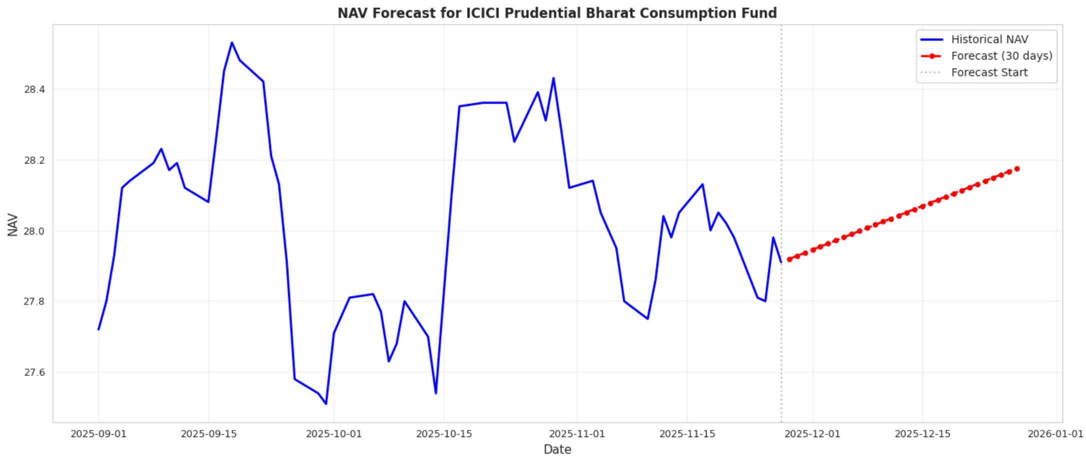

Figure 5.5: NAV Forecast for *ICICI Prudential Bharat Consumption Fund*. Historical NAV values are shown in blue, and the 30-day forecast is indicated in red with dashed lines and markers. The vertical gray line marks the start of the forecast period.

Observations

- The simple average-return forecast indicates a mild upward trend in NAV for the next 30 days.

- This approach assumes constant expected returns and does not account for volatility clustering or regime shifts.
- The forecast can be enhanced using advanced time-series models such as ARIMA, Prophet, or LSTM for more accurate and dynamic predictions.

Chapter 6. Conclusion & future scope

6.1 Findings/observations

The seven-year exploratory analysis of India's Consumption Thematic Funds reveals that while several schemes deliver strong long-term CAGR and competitive risk-adjusted returns, the category as a whole exhibits **high correlation** and **limited diversification potential**, as shown in the correlation matrix. Risk metrics such as volatility were relatively uniform across schemes, but downside metrics like maximum drawdown, Sharpe, and Sortino ratios displayed **large dispersion**, indicating substantial variation in downside protection. CAPM analysis demonstrated that **no fund generated statistically significant alpha**, with most schemes clustering around beta 1 and high R^2 values, suggesting closet-indexing behaviour. AUM-performance analysis further showed that fund size has **no meaningful relationship** with returns or skill.

Overall, the thematic category is dominated by market-driven performance rather than persistent manager skill, with only a few funds demonstrating consistently favourable Sortino-based rankings.

6.2 Challenges

The primary challenge was obtaining reliable long-horizon daily data across all consumption-thematic schemes. AMFI's interface only allows date-by-date retrieval, which made manual downloading impractical, as noted in the data collection section. This required building a custom scraping pipeline by reverse-engineering the AMFI API and handling issues such as API timeouts, intermittent failures, and the need for incremental saving to avoid data loss. Incomplete or inconsistent historical fields also created gaps that necessitated forward-fill imputation and careful cleaning. Collectively, these factors made the acquisition and preparation of a robust, multi-year panel dataset significantly challenging but ultimately manageable.

6.3 Future plan

Future extensions of this project could focus on expanding the dataset to include more schemes, longer time horizons, and additional market factors to build more accurate forecasting models. With richer data, the project can incorporate multi-factor models (e.g., Fama-French, macroeconomic factors), enabling deeper attribution of performance beyond CAPM. Incorporating fund-level holdings data would allow portfolio-level decomposition of sector exposures, risk contributions, and true active share. Machine-learning extensions could enhance prediction of volatility, drawdowns, and alpha using alternative data sources (corporate earnings, sentiment indices, macro indicators). Finally, building a live automated pipeline that updates NAV, AUM, and derived metrics daily would enable real-time monitoring and a dynamic composite scoring system that evolves with market conditions.

Group Contribution

Akshada Modak

- Worked on the data downloading
- Worked on the code
- Worked on the report
- Worked on the major part of the presentation

Maanav Gurubaxani

- Worked on the data downloading
- Worked on the data scraping
- Worked on the code for the analysis
- Worked on the major part of the report

Khanak Patel

- Worked on the data downloading
- Worked on the data scraping
- Worked on the initial analysis
- Worked on the initial part of the report

Short Bio

1. **Akshada Modak** is a B.Tech (Honors) student in Information and Communication Technology with a minor in Computational Science. She has strong technical experience in machine learning, deep learning, and database systems, having developed systems such as a disaster response coordination hub and an IMDb rating predictor. Her skills span Python, C++, SQL, and modern AI frameworks including BERT, TensorFlow, and Keras. Akshada has also architected BCNF-compliant relational schemas and implemented advanced SQL procedures for automation and data integrity. She is an active contributor to campus activities through the Placement Cell and Debating Society and has demonstrated strong analytical capability through national-level rankings and competitive programming achievements.

2. **Maanav Gurubaxani** is pursuing a B.Tech (Honors) in Information and Communication Technology with a minor in Computational Science. His technical strengths lie in systems design, algorithms, and database management, reflected in projects such as the Green Coin Management System and an optimized entry queue manager using C++ and GUI design. He has experience normalizing large database systems to BCNF and building efficient queue allocation algorithms that improve performance metrics. Maanav is skilled in C++, C, PostgreSQL, and related developer tools, and has demonstrated strong problem-solving ability through competitive programming and impressive national examination rankings. He is also actively involved in extracurricular initiatives, including dance and event coordination.

3. **Khanak Patel** is a B.Tech student in Mathematics and Computing with interests spanning AI/ML, computational social science, and full-stack development. She has worked on data-driven systems such as an investment management platform with a BCNF-normalized PostgreSQL backend and Flask APIs, as well as a Java-based railway management system using OOP principles and efficient data handling. Skilled in C++, Python, Java, SQL, and modern analytics libraries, she combines strong technical ability with experience in leadership and academic governance through roles in the Student Body Government and Debating Society. Her achievements include NTSE Stage 2 qualification, a 1530 SAT score, and selection for ACM India's Annual Event as a top Winter School performer.

References

[1] Classroom Discussion and Notes.

[2] AI used for finer understanding.
URL: https://gemini.google.com/
URL: https://chatgpt.com/

[3] AMFI (Association of Mutual Funds in India) webpage.
URL: https://www.amfiindia.com/otherdata/fund-performance